MALMÖ UNIVERSITY

Shaping Urban Living: Integrating Emotional and Speculative Design for Sustainable, Multifunctional Furniture.

A user centric design study for exploring possible ways to designing for future generations.

Forma urbant stadsliv: Integrering av emotionell och spekulativ design för hållbara, multifunktionella möbler.

En användarcentrerad designstudie för design för framtida generationer

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Abstract

The purpose of this study is to contribute to the design field with increased knowledge on how to leverage the emotional design perspective when designing for a future scenario set in Copenhagen in the year 2050. The research question is: How can product design, through emotional and speculative design, be used to create sustainable and multifunctional furniture tailored to the needs of future young adults in Copenhagen in 2050? The product chosen as the starting point is IKEA's stool "KYRRE." Through desk research, an analysis of the product's current design was created. A professionally developed future scenario for Copenhagen as a "15-minute city" in 2050 was then used as the basis for redesigning the "KYRRE" stool to meet the needs of future young adults. These new needs were utilized during idea generation and sketch development. A prototype was developed for testing in role-playing sessions where participants were briefed on the future scenario. The final design proposal, inspired by IKEA's "KYRRE" stool, satisfies the emotional, functional, and cognitive needs of a future urban lifestyle, focusing on sustainability, minimalism, and meaningful ownership. The stool is foldable, portable, and designed with attachable and detachable personal patches that function as digital keys, stimulating deeper social interaction. The stool becomes more than just furniture, it fosters social connections, self-expression, and sustainability, creating a stronger emotional bond with the stool. To design relevant products for a future sustainable and minimalistic society, greater emphasis should be placed on emotional needs. Embedded technology can be a cost-effective way to achieve this.

Sammanfattning

Syftet med denna studie är att bidraga med ökad kunskap till designkåren kring hur man drar nytta av det emotionella designperspektivet vid design för ett framtidsscenario som utspelar sig i Köpenhamn år 2050. Frågeställningen är; Hur kan produktdesign, genom emotionell och spekulativ design, användas för att skapa hållbara och multifunktionella möbler anpassade efter behov hos framtidens unga vuxna i Köpenhamn år 2050. Produkten som valts som utgångspunkt är IKEAs pall ”KYRRE”. Genom desk research skapades en nulägesbild av produktens design. Därefter användes ett professionellt utvecklat framtidsscenario för Köpenhamn som en ”15-minute city” år 2050 som underlag då pallen ”KYRRE” designades om för att passa framtidens unga vuxna.
De nya behoven användes vid ide- och skissutveckling. Prototyp utvecklades för test i rollspel där deltagarna briefades om framtidsscenariot. Det slutgiltiga designförslaget som inspirerades av IKEAs pall ”KYRRE” tillfredsställer de emotionella, funktionella och kognitiva behov som en framtida urban livsstil har, vari fokus ligger på hållbarhet, minimalism och meningsfullt ägarskap. Pallen är hopfällbar, bärbar och har designats med personliga på och avtagbara märken som fungerar som personliga digitala nycklar som stimulerar djupare social interaktion. Pallen blir mer än en möbel, den främjar sociala kontakter, sätt att uttrycka sig och är samtidigt hållbar, vilket skapar ett starkare emotionellt band till pallen. För att skapa relevant produktdesign i ett framtida hållbart och minimalistiskt samhälle, så bör ett ökat designfokus läggas på emotionella behov. Inbyggd teknologi kan vara ett kostnadseffektivt sätt att göra detta på.

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1.Introduction

-The need for smart solutions is huge when living spaces are minimal.
(IKEA Museum, 2023)

Today 4.4 billion inhabitants live in cities and are estimated to grow and double in size by the year 2050 (The World Bank, 2023).
Pavel Marceux, a managing editor at the AIM group, writes in his Article, A Look at the Future of the Family, that a larger number of singletons and childless couples are moving to megacities like London, Beijing and Tokyo looking for greater convenience and access to amenities and careers in urban areas (Euromonitor International, 2019).These smaller households are creating a boom in apartments that are tailored to one or two inhabitants (Euromonitor International, 2019). As a result of this, companies will see an increase in demands for products such as flatpacked furniture and foldable goods as well as other products that cater to smaller spaces (Euromonitor International, 2019). Thomas Armstrong writes in his book, The Human Odyssey: Navigation the Twelve Stages of Life (2019), that during early adulthood which include adults between the ages 20-35 the focus is to find a home, establish a family and friends as well as getting a job.

Filip Hoffman-Häuß (2023) writes in his article, Building a Sustainable Future: The Imperative of Creating a Sustainable Home, that the sustainable home is emerging as a crucial element in addressing resource scarcity and mitigating its negative impacts as well as climate change on consumers' quality of life (Euromonitor International 2023). He further explains that homes must evolve to integrate recycling, recovery and reuse across the entire lifecycle of products consumed (Figure 1) (Euromonitor International, 2023). Hoffman-Häuß writes (2023) that to be part of the sustainable future it's important that more strict laws are implemented, a change in consumer preferences, and to handle the rising costs for both companies and consumers which he says are driving the imperative.

As mentioned above, there is and will be a need for products that are designed and suited for smaller living spaces as well as a need for smarter and more sustainable furniture in the future. This study will examine how we today can design for these said future needs through one future design scenario.

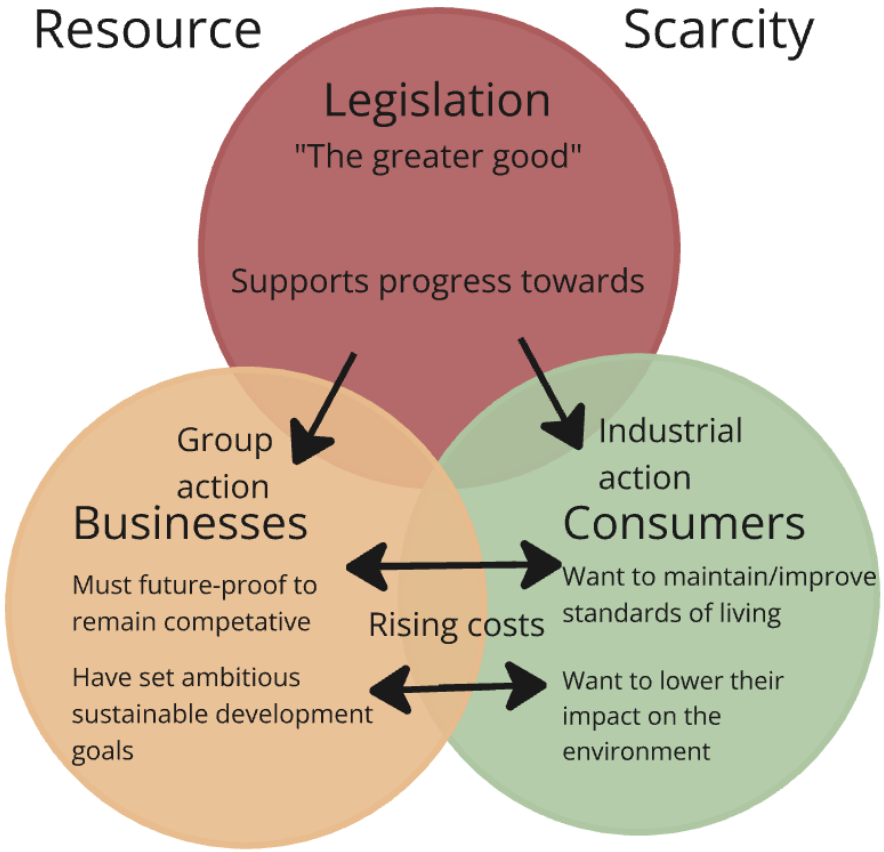


Figure 1: Shows how to build a sustainable future (Myhrman,2024). (Illustration developed from model by Euromonitor International (2023)).

1.1. Purpose of study

The purpose of the study is to contribute with increased knowledge for the design field on how to utilize emotional design perspectives for a future scenario taking place in Copenhagen year 2050.

A design contribution will be presented in the form of an experience prototype and will be based on the current existing stool “KYRRE” designed by IKEA.

A prototype was the chosen design contribution because it will showcase what needs and challenges are most relevant to focus on in the future.

The knowledge contribution will be new research questions and discussions that will arise from the prototype’s design features such as embedded technology and challenges with designing based on the future scenario.

1.2. Problem statement

How can product design, including speculative design and emotional design, be employed to create sustainable, multifunctional furniture tailored to the needs of future urban young adults in Copenhagen, year 2050?

1.3. Delimitations

In this study, it is essential to acknowledge and discuss certain limitations.

Ikea is aiming to be a multi international company but due to pressing time constraints this study will only use regional contacts. There will therefore be a geographical limitation with a focus on individuals aged 20-29 in Copenhagen.

The decision of the chosen target group stems from the observation that many people in this age range move into their first apartment and purchase their initial pieces of furniture. Expanding the target group further might compromise the depth of the data, rendering it too superficial for meaningful analysis.

The population of Copenhagen, Denmark in 2024 is approximately 1,39 milion people where as 11% are between the ages of 20-29 years old, which is estimated to be around 155 thousand people (Statistics Denmark, 2024).

The study will be limited to stools, as this furniture type holds the potential for meaningful exploration.

The study will not delve into materiality since it’s been predetermined that the end product will be constructed from wood. This limitation was imposed to ensure that the primary focus remains on sustainability and user-centric design. The same goes with ergonomics.

1.4. Ethical considerations

This study will not be tailored to people with disabilities due to pressing time constraints.

This study will have a sustainability focus in relation to the design. The material is predetermined and will not be changed even though it would benefit the sustainability aspect. Reason for this is pressing time constraints.

Since this study will involve human participants in one of the methods, it´s important that the participants personal data will be handled with care. The data collected will be managed in accordance with GDPR rules and guidelines as well as with the participants consent (Malmö Universitet, 2023)

2. Background

Three themes compromise the background:

- A future scenario which will play the part of the future aspect of this study.
- Information about an existing multifunctional stool and how its being used in an urban setting today as a starting point for this study.
- Relevant works in a creative manner that have influenced the study's design and progress.

2.1. New Urban Purpose- Copenhagen Institute for Future Studies (2022)

This report explores how our urban future could look like in Copenhagen by the year 2050 (Figure 2)(Copenhagen Institute for Future Studies [CIFS], 2022). They explore this through three key questions: What will make future cities attractive places to live, are generative cities possible and how can urban governance and leadership respond to the global challenges of the 21st century? (CIFS, 2022). This report does focus on Copenhagen for their future scenarios but also includes other geographical areas when presenting the information.

Established in 1969 by Professor Thorkil Kristensen, the OECD Secretary-General and former Finance Minister, the Copenhagen Institute for Future Studies is an autonomous, international, non-profit organization. It was founded in cooperation with forward-thinking Danish public and commercial institutions with the goal of enhancing decision-making via further research for the benefit of society. The institution wants to empower and encourage people, groups, policymakers, and the general public to start acting now on issues that will affect future trends and challenges.

This specific future scenario was chosen due to it being highly scientific and containing reliable sources, as well as presenting scenarios from both 2035 and 2050. Furthermore one of the future scenarios (year 2050) main focus was on sustainability and on its users. The future scenario also contains a lot of different aspects of how the future could be.

Further information about the future scenario will be explained under the method chapter, Future Design Toolkit.



Figure 2: Image of the scenario booklet (CIFS, 2022).

2.2. “KYRRE” by IKEA

“KYRRE” is a stool designed by Ikea with a minimalistic and multi-purpose design in mind (Figure 3) (IKEA, 2024a). Ikea describes it as “the savior when unexpected guests show up or if you need extra space or a book or drink” as well as “Stackable chairs so that you can have chairs for guests without taking up too much space”. Ikea also promotes the “KYRRE” stool on their website where they explain that a lack of space doesn’t have to mean that you can’t have room for a lot of guests.



Figure 3: “KYRRE” stool (IKEA, 2024a).



Figure 4: Seat cushion for “KYRRE”(Sikori, 2024).



Figure 5: Backrest for “KYRRE”(b.arc, 2023).



Figure 6: Hacking existing “KYRRE” (Fun Ways to Hack and Use the IKEA KYRRE stool - IKEA Hackers, 2020).

2.2.1. Hacks and personalization

Upcycling and hacking Ikea furniture is a popular thing on the internet. There are a lot of people who personalize their “KYRRE” stool by embroidery, painting, adding and taking it apart. One example of personalization comes from a Japanese company who creates seat cushions for the “KYRRE” stool and other already existing furniture (Figure 4) (Sikiori, 2024). Another example of hacks and personalization has been done by a Korean company called b.arc that designed a backrest that can be attached to the existing “KYRRE” stool (Figure 5) (b.arc, 2023). Users of the “KYRRE” stool have chosen to personalize the stool by painting it or changing the height of the legs as you can see from this user who posted on Pinterest (Figure 6) (Fun Ways to Hack and Use the IKEA KYRRE stool - IKEA Hackers, 2020).

These personalization and hacks goes hand in hand with Ikea’s home report for the year 2023 as well as how they predict 2024 trends to be colorful and include personalization and individuality (IKEA, 2024b), (IKEA, 2024c).

2.2.2. Democratic design: IKEAs design philosophy

The design of the “KYRRE” stool is in line with IKEAs design philosophy of democratic design, that has 5 different dimensions; function, form, quality, sustainability, and low price (IKEA, n.d.). With each principle considered, a design should be considered democratic (IKEA, n.d.).

2.3. Existing multipurpose furniture

2.3.1. Anna Kraitz - Cinderella Furniture of the Year (2023)

The winner of Sköna hems Furniture of the Year 2023 went to Cinderella, a multifunctional coffee table designed by Anna Kraitz (Designhousestockholm, 2023). Anna Kraitz said that her inspiration came from the ironing board that was kept away in a cupboard in her house taking up a lot of space. She recognized that maybe this hidden away ironing board could be turned into a more useful and aesthetically pleasing furniture that could be used in everyday life. She continued to say that she wanted to contribute with a design that followed the compact living trend where a lot of people want to live in the city center in small apartments with a lack of storage. She also said that for her it's a given to design furniture with multiple purposes and still be pretty enough to be displayed in the house.

The Cinderella coffee table could be used (Figure 7) as a stand up work desk, a coffee table, a side table for flower arrangements or a just stored clasped against a wall (Designhousestockholm, 2023). This design will be used as inspiration for this study for how to create new multifunctional furniture from already existing furniture.



Figure 7: Cinderella a multifunctional coffee table (Designhousestockholm, 2023).



Figure 8 and 9: Stool with a table with removable tray and pillows (Danyang, 2016).

2.3.2. Fika living: A Sharing Multifunctional Furniture for Single Person Households

In Chen Danyang masters thesis Fika Living (2016) she aims to create a sharing furniture for young single person households that will be used in the living room. Furthermore it was designed with the Swedish social phenomena "fika" in mind (Danyang, 2016). Danyang also had great focus on the sustainable aspect when designing. She chose to work with new environmentally friendly materials instead of using traditional materials only (Danyang, 2016). Furthermore she chose to include three aspects of the functionality of furniture which were, furniture structure, multifunctional and disassembly. Danyang also focused on how to satisfy the needs of the user.

The final design (Figure 8) (Figure 9) resulted in a stool with a table with a removable tray as well as removable pillows.

This design study has taken inspiration from this thesis by focusing on single person households but also including the social aspect as well as designing for the human needs.

This thesis includes aspects of tinyliving, multifunctionality and sustainability which are key factors in this study.

2.3.3. IKEA Big Ideas on Small Spaces - Existence Maximum

Ikea recently (15 December 2022- 3 September 2023) had an exhibition displaying prototypes from their own collection called Existence Maximum- Big ideas on small spaces (IKEA Museum, 2023). This collection of prototypes are multifunctional products designed for small spaces (IKEA Museum, 2023). The exhibition includes 14 rooms with solutions for widely different purposes. The rooms can also change character depending on need and activity (Figure 10), (Figure 11), (Figure 12). This is an example of how our future living situations might look according to Ikea which were used as inspiration in this study.

“One important aspect of our job is to look into the future, where we no longer think of the home as large or small, but instead focus on how much life it accommodates.”

- Anna Sandberg Falk, Curator, IKEA Museum (IKEA Museum, 2023)



Figure 10: Bedroom setup at the exhibition (IKEA Museum, 2023).



IFigure 11: Hallway setup at the exhibition (IKEA Museum, 2023).



Figure 12: Multifunctional table setup at the exhibition (IKEA Museum, 2023).

3. Theory

Research through design, utilizing preparatory studies, evaluation, and design work, serves as the foundation for the thesis (Löwgren, 2015). The goal of this approach is to advance scientific understanding by design (Löwgren, 2015). It is organized around the idea of exploring possibilities through a variety of design approaches (Löwgren, 2015).

In the following theory section, emotional design with a speculative design approach will be described. Emotional design will include the user-centric aspect of this study. To understand what the emotional needs are for the future generation, the speculative approach will be used to see how we can help design for a sustainable future. This studys main focus will be on the emotional bond to the stool by satisfying the emotional needs. These theories will interact to achieve the study's purpose.

3.1. Emotional design, designing for emotion

Emotional design or designing for emotion, is a design perspective where the leading principle in the design process is the intended emotional impact (van Boeijen et al., 2020).

When designing for emotion and for user-centric design, it's important to define the needs of the user, ideally it is what drives the development process (Hestad, 2013). As noted by Hedstad (2013), satisfying a need is the reasoning why we buy into a concept. Furthermore Hestad (2013) explains that a product can have the ability to satisfy a range of needs. The author (2013) further notes that an intended emotional impact can be a way to satisfy a need.

This following paragraph explains what different needs designers can design for and how to do it.

3.1.1. Human needs

One way to explain how to satisfy a need is the necessity to remove a defect from a product, solution, or situation (Wikberg Nilsson et al., 2021). To be able to create a functional, safe and dependable solution, you usually have to find a way to satisfy multiple needs, not only the physical. To create a more effective design other types of needs should also be concidered (Table 1).

How to create effective design that meet the different types of needs according to Wikberg Nilsson (2021):

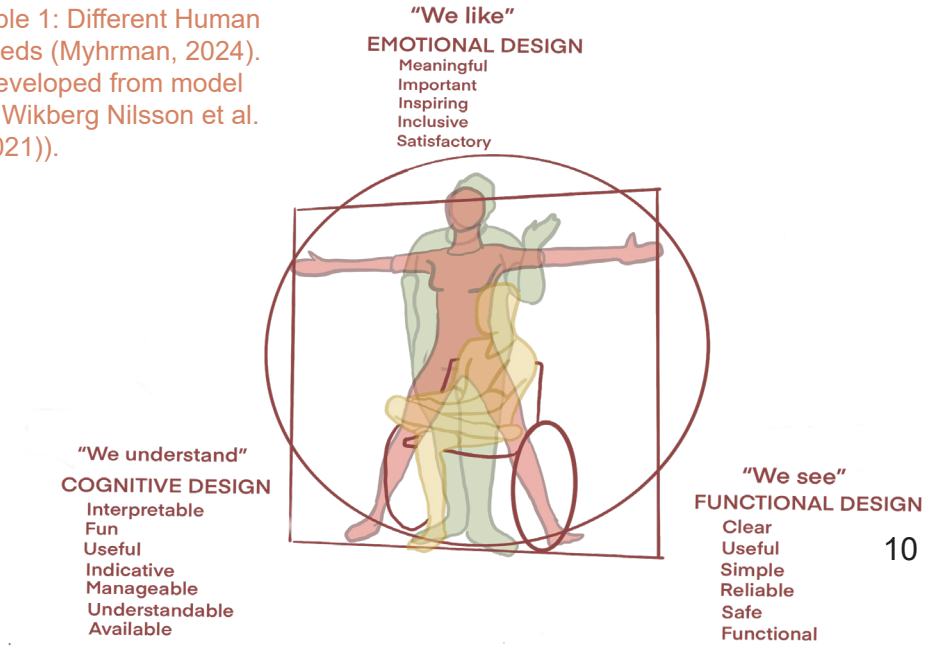
- **Functional Design should satisfy Physical needs**: entails improving physical features that are considered defective for the solution, situation or product.
- **Cognitive Design should satisfy Cognitive needs:** Another need is the cognitive needs which entails that the solution is comprehensible and practical for the user.
- **Emotional Design should satisfy Emotional needs**: Meeting emotional needs entail creating meaningful and enhancing certain user experiences - solutions that consumers like, ideally this would be because they gave the user that extra "wow" factor. This can then satisfy the emotional need in the design

It is important for designers to meet the user where they will be using the product or service to determine *implicit* needs and *explicit* needs. Explicit needs are needs we can express and identify. Implicit needs on the other hand are trickier to identify as the user or designer might not know they have those needs yet.

So to create an effective design, multiple of these needs are often met and considered.

This study's focus is to satisfy the emotional, cognitive and physical needs to create a more effective design.

Table 1: Different Human Needs (Myhrman, 2024). (Developed from model by Wikberg Nilsson et al. (2021)).



Needs according to: existential categories / Axiological categories	Being	Having	Doing	Interacting
Subsistence	Physical health, mental health, equilibrium, sense of humour, adaptability	Food, shelter, work	Feed, procreate, rest, work	Living environment, social setting
Protection	Care, adaptablility, autonomy, equilibrium, solidarity	Insurance systems, savings, cosial security, health systems, rights, family, work	Co-operate, prevent, plan, take car of, cure, help	Living space, social environment, dwelling
Affection	Self-esteem, solidarity, respect, tolerance, generosity, passion, sense of humour...	Friendship, family, partnership, relation-ships with nature	Make love, caress, express emotions, share, take care of, cultivate, appreciate	Privacy, intimacy, home, spaces of togetherness
Understanding	Critical conscience receptiveness, curios-ity, astonishment, rationality...	Literature, teatchers, method, educational policies, communica-tion policies	Investigate, study, experiment, educate, analyse, meditate	Settings of formative interaction, schools, academies, communi-ties, family...
Participation	Adaptability, recpetive-ness, willingess, deter-mination, dedication, respect...	Rights, responsibilities, duties, privileges, work	Become affiliated, co-operate, propose, share, interact, express opinions...	Settings of participa-tive interaction, parties, associations, commu-nities...
Leisure	Curiosity, receptive-ness, imagination, recklessnes, sense of humour, sensuality...	Games, spectacles, clubs, parties, peace of mind	Day-dream, brood, dream, remember, give way to fantasies, have fun, play...	Privacy, intimacy, spaces of closeness, free time, surround-ings, landscapes
Creation	Passion, determination, intuition, imagination, boldness, rationality, curiosity...	Abilities, skills, method, work	Work, invent, build, design, compose, interpret	Productive settings, workshops, cultural groups, spaces for expression...
Identity	Sense of belonging, concistency, self-esteem, differentia-tion...	Symbols, language, religions, habits, values, norms, memory, work...	Commit oneself, inte-grate onself, confront, decide on, recognize oneself...	Social rhytms, every-day settings, matura-tion stages...
Freedom	Autonomy, self-esteem, determination, pas-sion, boldness, toler-ance...	Equal rights	Dissent, choose, be different from, run risks, disobey...	Temporal/spatial plas-ticity

Table 2: Matrix over different human needs and their satisfiers (Myhrman, 2024). (Developed from table by Max Neef (1991)).



Table 3: Maslow´s Hierarchy of needs (Myhrman, 2024). (Developed from model by Maslow (1943)).

This paragraph explains more profoundly what different human needs to satisfy to live sustainably and this has helped to specify what needs and actions might be relevant later in this study (Max-Neef, M.A, 1991). These needs were classified by Max-Neef, Elizalde and Hopen-hayd in 1991 into axiological categories. Out of these needs they created a matrix (Table 2) which shows the different needs we people have and how they can be satisfied. The matrix shows examples of prospective satisfiers that could work across cultural boundaries.

The focus of this study has been on the axiological categories: *Identity* and *Affection*.

This study will also look into Maslow´s hierarchy of needs. In 1943 Maslow put out the theory of the hierarchy of needs (Table 3) (Maslow, 1943). These were basic needs we humans have for survival and include es-sentials like, food, water, shelter, air and warmth. These needs are reffered to as the human physiological needs.

A place to live is essential for human survival. As a place of refuge, a home might be considered the starting point for adressing specific needs.

The focus of this study has been on the tiers: *Self- Esteem Needs* and *Love and Belongingness* Needs, but might also explore Self Actualis-ation tier as well depending on what direction this study will go.

3.1.2. Emotional design and future sustainable consumption

Currently consumption is growing in magnitude while the throughput of manufactured goods is also developing at a rapid pace (Chapman, 2004). This causes an urgent situation because if no action is taken to reduce emissions, the global average temperature will rise over 2°C as early as year 2035 which will have detrimental effects on our planet (Stern, N., 2006).

Dr. Tim Cooper (2002) in his report "Durable Consumption: Reflections on Product Life Cycles and the Throwaway Society" argues that if we are to create sustainable consumption (Table 4) it requires products to have increased product life spans and demands so there is reduction in the throughput of resources (Cooper, 2002). Furthermore, he argues that if we want to go from the current throwaway culture to a culture where values are required for sustainability, we have to discover how to enable people to desire fewer material possessions while also forging *attachment* to those possessions already owned (Cooper, 2002).

The throwaway society still prevails, but there is a sign of an emerging culture critique of consumption patterns, which are often characterized by speed and short-sightedness (Cooper, 2002).

This study will focus on contributing to a sustainable future by exploring how to get a user to form *attachment* to a design for sustainable consumption.

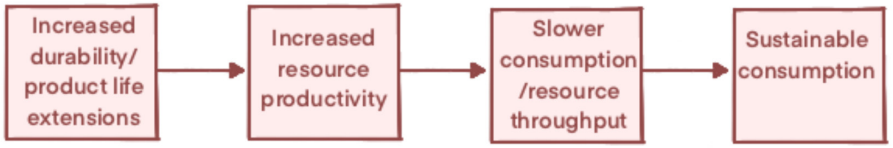


Table 4: Impact of product life on sustainable consumption (Myhrman, 2024). (Developed from Model by Cooper (2002)).

3.1.3. Studies about attachment and minimalism

The following paragraphs contains examples of studies that delves into how designers can help users form attachments to possessions already owned as well as how to enable users to desire fewer material possessions.

A case study was done to determine what causes attachments to products and what happens over time (Lee Crossley, 2004). The result of the case study showed that memories could significantly enhance consumer-product attachment (Lee Crossley, 2004). Furthermore, the study delves into what the determinants of the length of ownership could be (Lee Crossley, 2004). The result suggests that *enjoyment* may be the reason why users are attached to new products, whereas *memories* may be the reason why people are attached to products that they have owned for a longer time (Lee Crossley, 2004). The study mentions that *usability* also could be a determinant for enhanced consumer-product attachment (Lee Crossley, 2004).

Jiyun Kang (2021) proposed *minimalism* as a deliberate paradigm shift in consumer behavior based on the principles of sustainable lifestyle.

According to the Swedish Academy's dictionary, the term minimalism is defined as; a direction within art, literature, and music that uses minimum of expressive forms (Svenska Akademiens Ordlista [SAOL], 2015).

Furthermore, to provide its operationalization and evidence of its effects on emotional well-being, two studies were conducted (Kang, 2021). The first study explored how minimalism is structured by synthesizing existing measures of relevant constructs as well as other potential indicators that could represent the concept and help formulate minimalism's operationalization (Kang, 2021). The result showed that the second-order model of minimalism confirms the hierarchical structure of minimalism, which consists of four behavioral representations: Self sufficiency, cautious shopping, longevity, and clutter removal (Kang, 2021).

In the second study, a structural model and test were developed to depict the effects on minimalism on negative emotions (depression) and on positive emotions (flourishing) (Kang, 2021). The second study's results indicated that minimalism enhances flourishing and at the same time alleviating depression (Kang, 2021).

These studies mentioned above were used as a starting point for this study (Kang, 2021). Speculative design adds a futuristic perspective to this emotional perspective, which provides guidelines on how to design for emotion in the year 2050 (Kang, 2021).

The focus of this study is to get future users to form attachments to the design created as well as leave a desire to own fewer material possessions, which will further contribute to sustainable consumption. If we are to design for future users and satisfy their needs, we have to first understand what their needs will be. This can be achieved by combining speculative and emotional design.

3.2. Speculative design

Speculative design will be used to show the possible needs and social challenges that future urban young adults might encounter in a designed stool (van Boeijen et al., 2020).

The following paragraph will contain information about what speculative design is and how it can be used.

Speculative design is situated between emerging science, technology and daily existence (Malpass, 2017). Speculative design use future scenarios to showcase how designed items might be used in different contexts. Speculative design makes connections between future trends and the scientific research (Malpass, 2017).

Dunne and Raby (2024) just like Malpass finds speculative design as a way for imagining different futures to encourage critical thinking, their main focus is however less on scientific research but more on difficult societal norms through design and sparking debate.

Speculative design, instead of focusing on showing optimistic or pessimistic visions of the future, focus on raising important considerations regarding the societal and the ethical consequences of new science and how design and products might influence it (Malpass, 2017).

Julian Bleecker (2022) complements what Malpass (2017) says through his concept ‘design fiction’ where he uses narrative and speculative artefacts to question how society will be shaped by future technologies. Furthermore he also highlights the role of storytelling in exploring these consequences.

By raising issues with present design decisions and their potential effects in the future, speculative design challenges us to rethink how today might shape tomorrow (Malpass, 2017).

It makes people think about how modern technology and science are influencing everyones day to day life and how this might influence the future (Malpass, 2017).

Speculative designers do not have end-user goods in mind when designing but instead want to convey their research through physical objects (Malpass, 2017). These physical objects often get displayed in public places and therefore designers can adress concerns and educate the audience about new facts, showing the human side of difficult events (Malpass, 2017). Speculative design helps to increase public awareness by providing workable solutions and critical analysis rather than easy fixes and impractical technologies (Malpass, 2017).

Dunne and Raby (2024), just like Malpass (2017), use their designed concepts as a way to get people to talk and challenge their assumptions. Dunne and Raby (2024) and Malpass (2024) therefore have a greater focus on public education through objects, while James Auger’s (2013) aim is to encourage reflection on how modern material culture can change over time by combining familiar everyday objects with future technologies in his designs.

Speculative designers base their creations and designs on the surroundings and on the things of the modern world (Malpass, 2017). The speculative design approach believes that the modern material culture will endure and that new and old things will coexist (Malpass, 2017). This can bee seen in the hypothetical scenarios that both have technology from the future blended with modern everyday goods (Malpass, 2017).

Speculative design embraces complexity (Malpass, 2017). This can be shown in the way that this approach illustrates how modern technology interacts with traditional artifacts and behaviors (Malpass, 2017). This can reveal problems that could potentially hinder the development of new technology (Malpass, 2017).

Speculative design produces alternative presents or simple futures through scenario writing and shows everyday life with certain changes to draw the attention to certain potential problems (Malpass, 2017).

4. Method

The method chapter is built up in three parts. The first is to understand the existing design of “KYRRE” stool and existing customers relations to it, with a focus on emotional design. This was done by doing secondary research, where guided questions were created by observing and interacting with the stool to make it easier to find and gather answers from existing sources. The second part is to visualize the future persona and scenario where the stool “KYRRE” is a relevant furniture. Here a professionally developed scenario was used to create the foundation for using a speculative design method. The future scenario and user were visualized based on a future design toolkit that included scenario presentation, “story world” and “day-in-a-life” presentation. The third and last part is the design process, where we redesigned the stool “KYRRE” to be suitable for the future scenario chosen and for the future persona. An experience mock-up prototype was developed and evaluated through role-playing with live actors living the future scenario.

4.1. Desk research / Secondary research

Because of technological advances, a lot of various data has been collected that is easily accessible at all times for research purposes (Johnston, 2014). This has led to secondary research becoming a flexible and popular method to use. Secondary research involves using information that was gathered for another primary research purpose. Secondary research is usually used when there are pressing time constraints for the execution of the research.

The method began with an investigation to learn what existing information was available and what remained to be learned or discovered. The first step when conducting secondary research was to create research questions for the particular area being investigated. This study already had a research question, but as it was too general, it was decided instead to create guided questions. These guided questions were created by observing, assembling, and interacting with the stools. These guided questions were also created to explore particular aspects of the stool, such as the sustainability aspect, the design, the usability, users and the relevance (Figure 13). The objective was to gather specific information about the stool that would deepen the understanding of its characteristics and help the study's goal of collecting contextual information.

This method was conducted in the preliminary phase of the study to help gather contextual information about the stool. For this method, two "KYRRE" stools were bought, assembled, and placed within eyesight for examination (Figure 14). For the secondary research, information was gathered from IKEA's own website, IKEA.se. The secondary research took place on the 12th of February 2024, lasted approximately three hours, and was conducted in an apartment.

First, the questions were written down using a computer while simultaneously observing the stools. Thereafter, the questions were used as a guide for finding more specific data on the IKEA website. The data was then collected and used to contribute to answering the study's research question.

Guided Question

1.What makes you different from IKEA's other stools? What was the purpose of designing you?

2.What makes you multifunctional? Was that your intended use or just coincidence?

3.For what target group were you desgined for? How does it show in the design and on the assortment?

4.Where and how is the intended use of this stool? Where in the apartment? How does it show?

5.How are you built and designed with IKEA's design philosophy in mind?

6.IKEA's sustainability view, 3 materials, are they all necessary?

7.Size, design and colour. What purpose do they have or is it just because of design purposes?

8.When was it released and how was it improved from your previous version?

Figure 13: Guided Questions for the desk research that was made by looking and interacting with the "KYRRE" stool (Myhrman, 2024).



Figure 14: Assembling the "KYRRE" stool (Myhrman, 2024).

4.2. Future design toolkit - written scenario

Future design toolkit is a number of methods that designers can use to design for complex tomorrows (Fuel4Design-Future Design Toolkit, 2021). The toolkit contains 8 general methods with 18 sub categories with more detailed methods that can be done (Fuel4Design- Future Design Toolkit, 2021). In this study three of the subcategories were used as methods and are: Scenario, A day in a life and Story world which are under the categories Scenario and Future Persona. The other methods meaning: Horizon scanning, Framing Signals, Future Philosophical Pills, Design Interventions, Mapping Consequences and Provotyping were not used because this study already had a general scenario as a starting point and did not need to create a scenario from scratch.

A written scenario is a method used by designers to tell a story about the intended user experience in a specific situation (van Boeijen et al., 2020). A future scenario can be used to determine the possible interactions, or possible interactions in a future situation that creates certain experiences (van Boeijen et al., 2020).

Executing this method helped to further understand what setting this design should tailor too. The emotional needs are the same but the setting is different which will significantly change the design outcome which is why its important to have a scenario to refer too.

This method took place on April 14, 2024. A premade template was used (Figure 15) from future design toolkit to help turn a premade scenario into a comprehensible scenario that was more suitable for when designing (Fuel4design, 2021). The information used for the template came from the prechosen scenario: New Urban Purposes (2022) written under the chapter background. No outside information was gathered for this method only information from the chosen scenario.

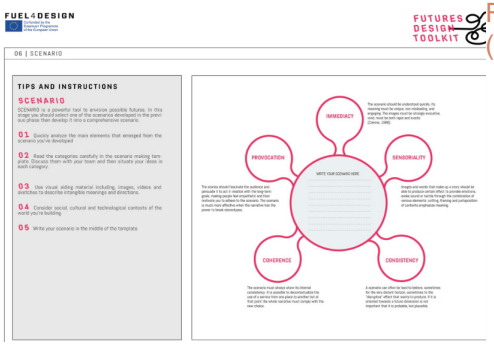


Figure 15: Future scenario template (Fuel4Design, 2021).

4.3. Future design toolkit - future persona

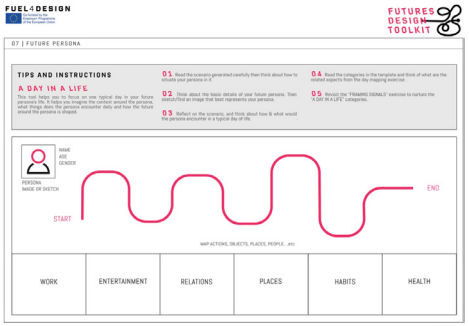
The design method future persona is used by designers when you want to empathize with the users you are designing for (van Boeijen et al., 2020). You do this by creating an archetypal representation of your intended user groups that visualizes and describes their needs, values and behavior based on insight into their lives (van Boeijen et al., 2020). These personas can then later help you to communicate and be aware of these real life needs, values and behaviors in your design work (van Boeijen et al., 2020). For this method two different templates were used from future design toolkit (Figure 16 & 17). This method was executed on April 14, 2024.

4.3.1. Story world

The first template, story world, was used to gather an overview of the characteristics including their needs, aspirations and other insights of the target group (van Boeijen et al., 2020). This can help to uncover hidden and unspoken aspects of the users life (Fuel4Design, 2021).

4.3.2. Day in the life

The second template, a day in a life, is a tool that is used to get a better understanding of what a typical day can look like for your user. This tool can be used to understand the context around the future persona, how the future around them is shared and what things they encounter daily (Fuel4Design, 2021) .



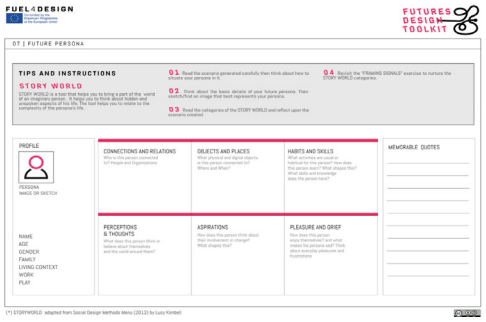


Figure 16 & 17: Future persona, two different templates (Fuel4Design, 2021)

4.4. Prototyping and design proposal

Prototyping is a great method to use when exploring the design problem and finding solutions for it (Wikberg Nilsson et al., 2021). In this study, prototyping played an important role, as it took the design from the initial idea generation to refined concepts.

4.4.1. Sketching

Sketching is a great initial method to start the design process, as it helps to develop and explore different design concepts (Wikberg Nilsson et al., 2021). Sketching is a way to quickly visualize ideas. Since the preliminary phase of this study, ten-minute sketching sessions have been conducted each day to generate new ideas or get valuable insights. This phase of the study is an effective way to generate different concepts without worrying about perfection. After finding a concept that works, more detailed sketches were created. When the detailed sketches proved effective, they were then turned into simple cardboard mock-ups. As insights were gathered from the mock-ups, improvements were identified, initiating another sketching phase. Thereafter, new mock-ups were created. In this study, sketches were closely related to the prototyping phase and were mostly done simultaneously.

4.4.2. Experience prototype

Experience prototyping was used in this study to get instant feedback from the target group and to test the design in real-life situations so that the design could be further improved (Wikberg Nilsson et al., 2021). These experience prototypes were made out of cardboard mock-ups. Cardboard mock-ups are easy to use, as they can easily be modified if needed but still convey the design concept for users to understand. Mock-ups are used to test the design's functionality and to observe how users interact with it. The users will interact with it through another method called role-play, where the target user engages with the design in an environment and manner similar to its intended use.

4.5. Role-play

The experience cardboard mock-up prototype can be used when conducting role-play to observe the area of use and to let the intended users test the design. It can also be used in experience prototyping by creating further insights to improve the design concept after role-play (Wikberg Nilsson et al., 2021).

Role-play is a design method that helps the designer simulate a possible user experience and to further understand the user's interaction with the design itself. The method can be used in multiple ways. It can either be used during the whole design process or as feedback with the future users. Role-play is a good method to use for evaluating the design from a user's perspective and to identify the design flaws and to gain new insights (van Boeijen et al., 2020).

Role-play was chosen as a method for this study because it seemed suitable for achieving the research objective, which is to determine the emotional value gained by the design and its interaction in the future of year 2050 in Copenhagen (van Boeijen et al., 2020).

The role-play with two participants took place on July the 18th, 2024 from 07:00 am to 12:00 am. The two participants were participant 1, 25 years old and participant 2, 56 years old. Participant 1 played the role as Jack from the provided scenario of Copenhagen 2050. Jack is a 25-year-old student using advanced AR/VR contact lenses and AI.

Participant 2 played the role of a stranger Jack met in a park and became acquainted with.

The role-play took place in the scenario previously mentioned where key services are accessible within 15 minutes within the city, ”15-min city”.

Before the participants arrived at the location the costumes and props were prepared which are; the prototypes, backpacks, coffee, cups, recording equipment such as iphone voice recorder, notebook and a pen.

The chosen location for the role-play was Malmö due to having geographical similarities to Copenhagen where the chosen scenario takes place. Two locations were used. Location 1 was on a street where Participant 1 was filmed walking down a longer street wearing the stool folded on his backpack. In the end he bought a coffee with his AR/VR lenses. Location 2 was be at a park where participant 1 sat down on the stool to have a coffee. Participant 2 appeared and sat next to Participant 1 on his stool and also had a coffee while they shared a conversation. They were provided with an example of how the dialogue could go in their paper booklet, but they were free to choose how the conversation could go. Participant 1 eventually left when his coffee was done and they would exchange patches (Personal Digital Keys) and then participant 1 scanned the patch he received from participant 2 to learn more about who he is. The patches included embedded technology, to access your personal network, and are part of the stool.

All of this was recorded both in video and voice recording as well as with photos and notes with a notebook and pen.

After the interaction the participants were asked questions about their experience (Appendix 7) that were used to evaluate the design further. The feedback given resulted in that new potential problems appeared and thereby sparked new important debates. This is the intention by using speculative design.

5. Results and analysis

In the following section the results and analysis from the executed method was presented. The results and the analysis will be interwoven to give a better overall perspective. From the method, executed design decisions were made, which was discussed more in depth in the following section. Ultimately, these design decisions were used to develop an experience mock-up prototype which was the design contribution of this study. For clarity this chapter follows the same structure as the previous method chapter.

5.1. Result and analysis of desk research

The desktop research resulted in valuable information about what to think about when designing a multifunctional sustainable furniture and IKEAs thought process behind it (Appendix 2). It also resulted in a starting point in where we are today when it comes to multifunctional furniture for tiny urban living.

Key insights gathered from desk research:

1.Multifunctionality: Designed to be multifunctional for compact living. Usually used as a side table or extra seating for guests.

2.Design Evolution and Sustainability: The seat of “KYRRE” was updated into a rounder triangular shape for economic and sustainable reasons. Cutting the shape more triangular created less waste (Figure 18 & 19).

3.User Feedback: Reviews showed majority of users being pleased with the design whith a minority saying that the assembly was difficult (Figure 20).

4.Customization: When analyzing todays design of IKEA’s stool “KYRRE” we found that it’s designed with a main focus on satisfying the physical and cognitive needs but is lacking in design features tailored to satisfy the emotional need aspect. According to Wikberg Nilsson you have to satisfy multiple of the needs to create an effective design (Wikberg Nilsson et al., 2021). To satisfy the emotional needs the design should according to Wikberg Nilsson, create meaningful and enhancing certain user experience solutions that consumers like. In short, a “wow” factor. (Wikberg Nilsson et al., 2021). Based on the result of the desk research and the theory, “KYRRE” could achieve a more effective design tailored to the users needs by also satisfying emotional needs.

We found in the IKEA reviews that some users had started to customize and personalize the standard product but was not driven by the producer IKEA. These end users are by own choice satisfying their emotional needs of enjoyment and creating memories with the stool.

5.IKEA’s Democratic Design Philosophy: “KYRRE” is a good example of IKEA’s commitment to creating sustainable, affordable and functional design for everyone.



Figure 18: “Mjölkpallen” designed by Ingvar Kamprad year 2004 (Auctionet, 2024).



Figure 19: “KYRRE” designed by Mickael Axelsson year 2019 (IKEA, 2024a).

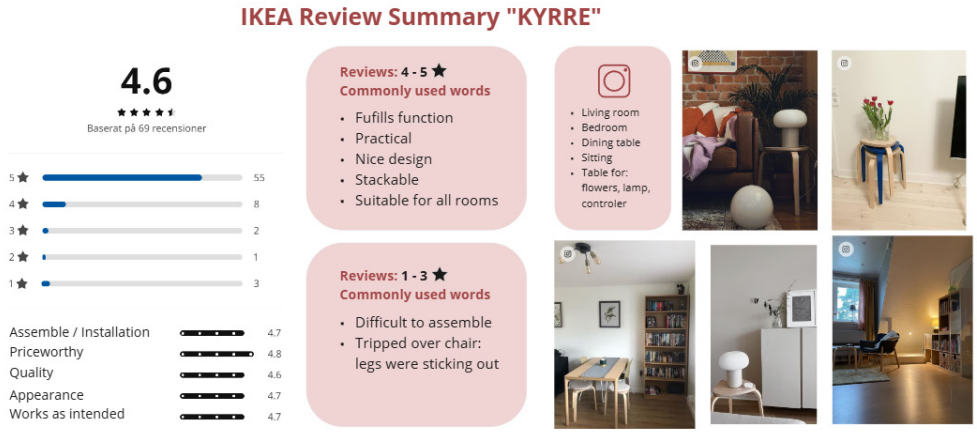


Figure 20: “KYRRE” IKEA review summary (Myhrman, 2024). (Developed from website by IKEA (2024)).

5.2.Result and analysis of future design toolkit - scenario 2050

5.2.1. Result from written scenario

The written scenario resulted in a more clear understanding of the setting this design would be designed for. The written scenario also helped to illustrate the technological changes coming our way and how they may impact urban spaces and daily lives of young adults, 20-29 years of age (Figure 21).

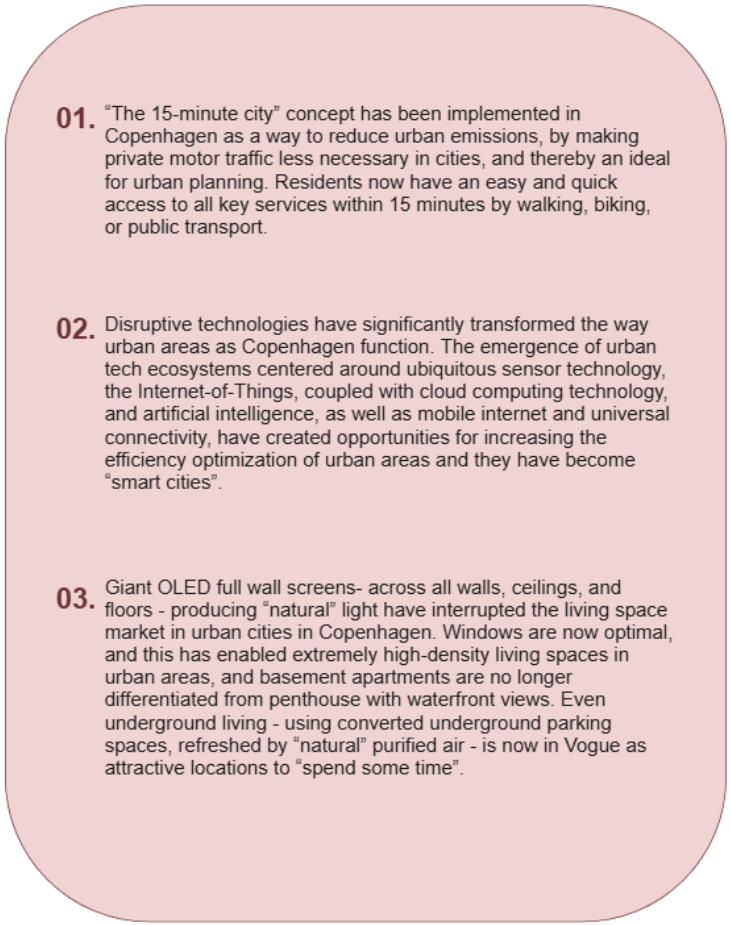


Figure 21: Result from future scenario template (Myhrman, 2024).

5.2.2. Analysis from written scenario

In the professionally developed scenario for year 2050 we are pushed out of our comfort zone and forced to think differently when redesigning everyday furniture as the stool "KYRRE". As mentioned by Malpass (2017), speculative design is using future scenarios to make connections between future trends and the scientific research as well as how designed items might be used in various future daily life situations (Appendix 3). Sustainability focus, away from "Throwaway society", has in the scenario driven minimalism and a new way of thinking about remaining possessions (Kang, 2021). According to a study made by Kang (2021) the results indicated that minimalism even enhances flourishing and reduced risk for depression. As mentioned by Tim Cooper (2002) we have to discover how to enable people to desire fewer material possessions while adding emotional attachment to possessions already owned. The scenario of Copenhagen year 2050 gives a new reference point for creating effective design where we have to incorporate new technology, minimalism and a greater focus on services rather than just focusing on the physical product itself.

5.3.Results and analysis of future design toolkit - future persona 2050

5.3.1. Results from story world

Jack is a 25-year-old student "living" in Copenhagen. He has universal connectivity via a distributed satellite-based global cloud network and uses augmented and virtual reality, voice-interactive control contact lenses to request recommendations with his AI-driven personal digital assistant (PDA), Johnny-Boy, for all aspects of his life. As a result, Jack maintains a fully on-demand lifestyle, embracing the sharing and circular economy more out of convenience and low cost rather than "green" motivations. Jack thinks about his consumption entirely within the everything-as-a-service (XaaS) space. He does not own or even rent an apartment, and only "lives" in Copenhagen when he chooses. He is able to locate living quarters or even just a bed for the night on a very short-notice basis and is never much concerned with where he might be in a day or two. Transportation, food, entertainment, and education are just as, if not more, convenient, and low-cost. Johnny-Boy monitors Jack's mood and creates hyper realistic, expensive, or immersive views, based on his understanding of Jacks' preferences. There is increasing hostility- especially among older generations, toward the tech-driven laissez-faire lifestyle of Jack and his generational cohort.

5.3.2. Results from a day in the life

- Traveling light with essentially only his backpack as his belonging.
- Sleeps at new locations every 2-3 days, which means traveling frequently.
- Have multiple social interactions per day with new people/old friends.
- Lives in 15-minute cities which means a majority of the day he is walking and/or taking public transport.

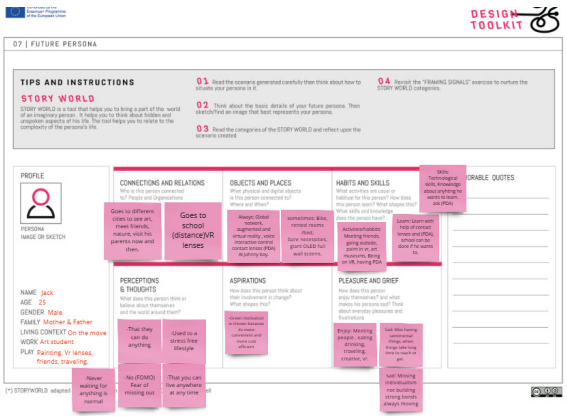


Figure 22: Results from future persona, Story World template (Myhrman, 2024). (Developed from template by Fuel4Design (2021)).

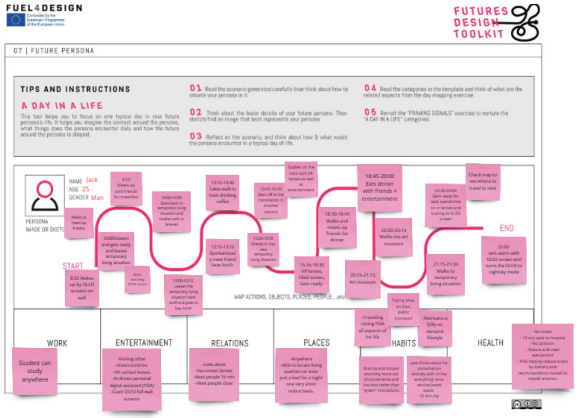


Figure 23: Results from future persona, A Day in a Life template (Myhrman, 2024). (Devloped from template by Fuel4Design (2021)).

5.3.3. Analysis future persona

The future persona method helped to understand the future user living in the future scenario. The insights were essential to be able to create user-centric design (See figure 22 & 23). Jack is continuously on the move and only possesses a few items that he can bring with him. To be able to qualify to be one of those items, the design must be effective and satisfy emotional needs in addition to physical and cognitive needs, which also is mentioned by Wikberg Nilsson (Wikberg Nilsson et al., 2021).

It can be seen that in a minimalistic society as presented in the scenario, the item would benefit from being multifunctional and ready to bring with you when moving to another city or living in an apartment.

In a case study Lee Crossley (2004) showed that memories could significantly enhance the consumer-product attachment and thereby satisfy the emotional need. *Enjoyment* may be the reason why users are attatched to a product and *memories* may be the reason for *attachment* to a product they have owned for a long time (Lee Crossley, 2004). In addition, *usability* could be a determinant for enhancing consumer-product attatchment (Lee Crossley, 2004).

In addition we learned from Future Persona that Jack is a young urban citizen with many social interactions with old and new friends every day. Based on Maslow's hierarchy of needs we assumed, based on the results, that Jack's Physiological and Safety & Security needs are met and that his focus is on Love & Belonging, where he puts focus on friendship, family, intimacy and connections (Maslow, 1943).

These are important factors to consider in our design process when redesigning the stool "KYRRE" to be a sustainable and multifunctional furniture tailored to meet the needs of future urban young adults living in the year 2050 scenario. For more details look at the results in (Appendix 4 & 5).

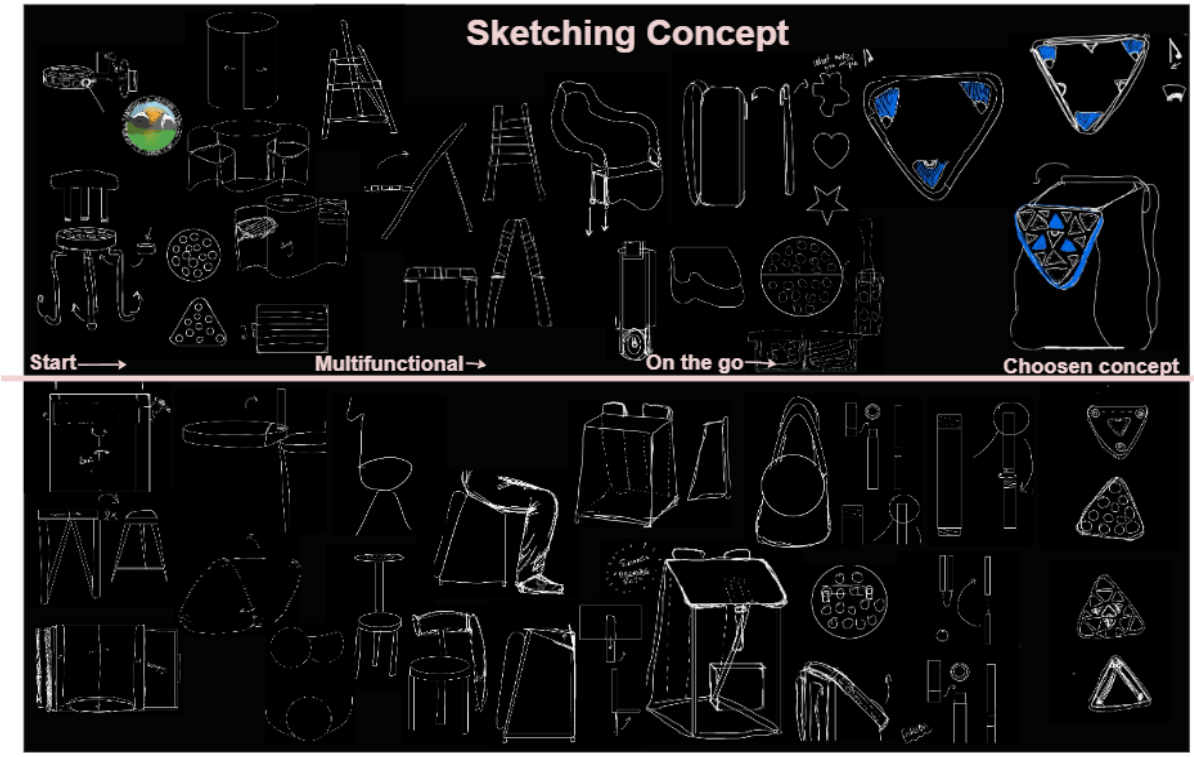
5.4. Results and analysis of prototyping and design proposal

5.4.1. Results from first round of sketching and cardboard prototyping

The results from the first round of prototyping were sketching prototypes (Figure 24). These prototype sketches were made during the whole design process to keep multiple design solutions in mind. They further developed when the “on the go” concept was introduced into the design process. This resulted in a smaller and more intricate design. Furthermore the concept of having it attached to the bag itself was introduced.

A more circular version of a stool was introduced in the earlier stages of sketching (Figure 25). This sketch was further explored by making a simple physical cardboard mock-up that was attached on a backpack (Figure 26).

Figure 24: Beginning stages of sketching (Myhrman, 2024).



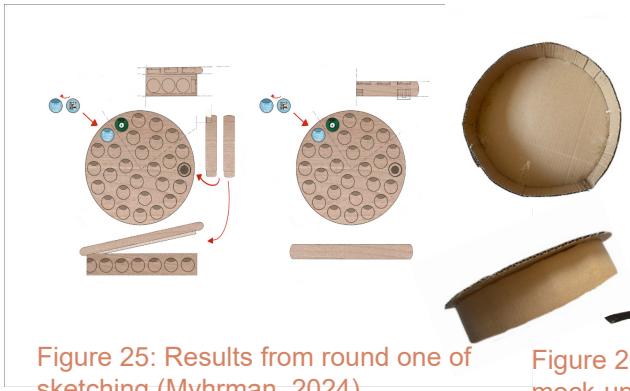


Figure 25: Results from round one of sketching (Myhrman, 2024).



Figure 26: Results from round one of mock-up prototyping (Myhrman, 2024).

It was decided that this concept would not be developed further due to it not being relevant for the user. The holes and building the stool by hand everywhere was determined to be inefficient and taking up too much of the user’s time who would have to ensemble it multiple times a day. Furthermore there were not enough needs for that many features of the stool. The circular shaped stool wasn’t space efficient enough and therefore not chosen to explore further. For more details about the user-centric design see appendix 8.

5.4.2. Results from second round of cardboard prototyping

This decision meant that more prototype sketching were done. This time the original shape of the “KYRRE” stool was explored due to it being space efficient. It was further decided that the multifunctionality would be the same as the original “KYRRE” stool with the added features of portability and customization with the interchangeable patches (Figure 27 & 28). This sketching was further developed and made into cardboard prototypes that was created with the same measurements as the original “KYRRE” stool. This prototype also included a feature to hang it on a backpack as well as stool legs that could be folded to the side for further convenience when carrying.



Figure 28: Results from round two of mock-up prototyping (Myhrman, 2024).



Figure 27: Placement of patches (Myhrman, 2024).

The patches were tested in different arrangements and shapes with sketches to determine the best fit for the design. The placement of the patches was determined by how many patches could fit as well as the aesthetic factor (FIgure 29). Furthermore different shapes were explored to best fit the triangular shape and be aesthetically pleasing (Figure 29). The patches were further developed through sketching to be more user friendly (Figure 29). The result was an ergonomically, esthetically pleasing and customizable patch that could be exchangeable during social interactions (Appendix 8) (Figure 29). To further enhance the emotional value of the patches, embedded technology was added. This technology can be used to quickly give acces to eachothers personal digital network and thereby stimulate interactions and building relationships. The name of the patches were changed to Personal Digital Keys. The prototype was further developed by being painted and thereby adding more realistic view on how the final design would look like (Figure 29).

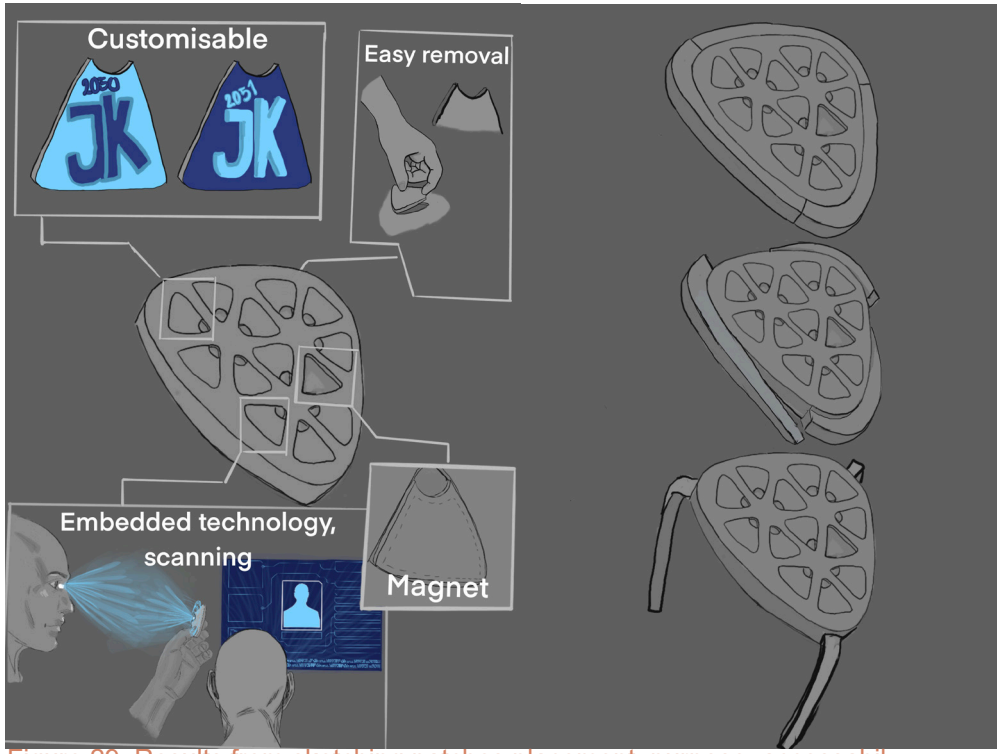


Figure 29: Results from sketching patches placement, purpose, accessability, customizability and how to unfold the stool legs (Myhrman, 2024).

IS-DO-VALUE Description of the final design concept

IS:

The new ”KYRRE” stool is designed to be:

- A stable construction in natural materials, wood
- Multifunctional, being both a stool and table
- Foldable to use less space
- Portable and easy to carry on a backpack
- Customizable with removable patches
- Using embedded technology creating access to personal digital networks

DO:

- The new stool makes on-the-go lifestyle easier, by being multifunctional, foldable and portable
- The new stool builds a persons unique identity by being customizable with different designs of patches
- The new stool makes it easier to connect and stay connected with new friends. The patches (personal digital keys) have embedded technology in them so that they can be exchanged with friends and scanned with VR lenses to get access to the persons information. This encourages social connections.

VALUE:

- The new stool design supports the young urban users on-the go lifestyle that is sustainable and minimalistic. This adds emotional value and attachment to the stool.
- The patches (personal digital keys) help create a sense of belonging and identity by giving the young urban users a place to share parts of who they are in a way that is easy to access and personal. This elevates the stool and turns it into something more than just normal furniture. It now invites connections between people, a conversation starter, and brings people closer to each other. By using embedded technology, the patches (personal digital keys), give access to a new personal digital world making it easier to connect with excisting and new friends. This creates instant engagement and makes interactions with people more meaningful and unique to everyone. Emotional attachment to the stool will be created through enjoyment and positive memories.

For more details about the user-centric design process, see appendix 8.

5.4.3. Results and analysis of prototyping and design proposal

As mentioned by Wikberg Nilsson (2021), sketching is the first step from taking an idea to a solution. By starting the sketching process early, new ideas were tested quickly and the iterative thinking and sketching process helped to further develop the design of “KYRRE”. Initially, before the scenario was introduced, the focus was on sustainable and minimalistic design only. The focus was on smaller changes more narrow in time, mainly physical and cognitive. The perspective changed a lot after the introduction of the scenario, and use of speculative design theory, The Scenario of Copenhagen year 2050 and “The 15 minute city” set the scene where the product must fit (CIFS, 2022). In addition the Future Persona painted a picture of the user and its needs. The focus shifted to make an effective design also including meeting the emotional needs (Wikberg Nilsson et al., 2021). It was found that the design would benefit from being multifunctional and foldable so that it was easy to carry with you. The emotional need could be met with attatchment that creates enjoyment and memories. It was determined that technology could benefit if added to the design.

The idea of creating patches (personal digital keys) to customize your stool was born. By using and wearing the foldable stool attatched on your back-pack, it shows the user, their friends and their friends-to-be that they are part of a network of people and open for conversation and connection. By exchanging the patches (personal digital keys) acess is given to the users personal record and in an instant the user gets ideas and inspiration on conversation topics and joined interests.This design feature could support Jack in fufilling the need of Belonging & Family (Maslow, 1943). To live sustainably, (Max-Neef, 1991), identified a number of need satisfiers and from this matrix the need satisfiers chosen were of Identity and Affection, most suitable for the future persona Jack. The design of the patches (personal digital keys) is clearly visible on the seat of the stool to be clearly visible to others and yourself, supporting the need satisfiers Identity and Affection (Figure 30). The end result was an experience cardboard mock-up prototype of the stool to be used in role-play for further insights (Figure 31).



Figure 30: Final design contribution in the form of a mock-up prototype (Myhrman, 2024).


Figure 31: Final design contribution in the form of a mock-up prototype, unfolded and folded (Myhrman, 2024).

5.5. Results and analysis of role-play

The role-play method resulted in some valuable insights about the design concept (Appendix 6 & 7). These insights were gathered through observing, interviewing and receiving feedback form the participants.

The experience cardboard mock-up prototype was used when conducting role-play to observe the area of use and to let the intended users test the design. It was also used in experience prototyping by creating further insights to improve the design concept after role-play (Wikberg Nilsson et al., 2021). A design concept was made to meet the needs of Jack, age 25, living in Copenhagen, year 2050. To incorporate a user-centric design perspective, it was decided to use role-play where participants were briefed of the assumptions of the year 2050 scenario and to use that perspective in their actions, discussions and reflections. The role-play worked as feedback on the design concept and helped to confirm the concept and to further identify areas of improvement. As mentioned by Malpass (2017), speculative design embraces complexity and illustrates how modern technology interacts with traditional artifacts and behaviors.

The two participants found it realistic that sustainability drives minimalism and that minimalism makes it possible to have more free time to meet up with people, have real conversations and relations creating positive memories (Appendix 6 & 7).

They found the stool easy to unfold as well as to exchange patches as the process is similar to today's tradition of exchanging business cards. The patches worked very well as conversation starters. It was possible to directly see what friends he has. It was considered cool that it was possible to scan the patches with your AR/VR lenses and that it provided conversational topics that could start directly. Sitting on the stool, covering the patches, were seen as a problem because when the user had to stand up and show the patches to the friend they had to get up. When using the stool as a table it worked much better. There was also an interest in using a lighter material than wood that were more futuristic and sustainable.

They found it easy to use the patches in conversations and really liked the idea of how two worlds quickly are connecting and that AI quickly can be used to find common topics to discuss.

The stool was easy to carry with you and use. They liked that the shape of the seat and the patches were the same. They also appreciated the foldability, going with ease from flat packed (2D) to an unfolded rigid stool (3D). Because the user don't own many things, it's important to have a conversation starter that makes it easy to share memories and get that positive homey feeling. It is so cool that it is possible to save and share memories with the patches. It was repeated that with minimalism conversation topics change from what you own to who you are. This creates more meaningful conversations and positive memories. When scanning the patches the user could see all the information about that person and quickly be given multiple topics to discuss. Conversations become much more effective and build on trust. A lot better than today's social media where, very little information needs to be known before befriending someone.

When asked if they would feel more for this new stool than excisting, the answer was yes. When meeting more and more people and collecting their stories on the stool it becomes more valuable. But then the chair doesn't become a chair, but also the people behind it. It almost bears similarity to a photo album where the persons's memories are stored.

How can the stool be improved? They wanted lighter material of a future sustainable material instead of heavy wood. Recycled aluminum was suggested and discussed. They wanted a stool that is light and invisible and that is developed together with a backpack that can carry the lighter stool. In addition the software supported by embedded technology needs to be developed further. Starting of using QR-codes and mobile phones to test it out and transition to somehting more futuristic tech later.

However, they felt a bit awkward that the stool was the most important piece of furniture when owning almost no other pieces of furniture. Seating areas can be found almost anywhere. There can be alternatives to a stool that fufills the same function as being a carrier of memories and conversation starter. A jacket with patches or a backpack with patches could be two alternatives that can be further explored.

The final prototype that was used in the role-play was developed based on human needs and insights developed during the prototyping and role play process. The further the design process developed the more focus were put on emotional design elements. The final role play confirmed that this was very important to them.

For more details about the user-centric design process, see Appendix 8.

6. Conclusion

The problem statement of the study was ”How can product design, including emotional and speculative design, be employed to create sustainable, multifunctional furniture tailored to the needs of future urban young adults year 2050?”

The study showed that emotional design and speculative design in combination could be used to evaluate and test existing designs for their relevance in the future. By studying the current design of the ”KYRRE” stool from an emotional design perspective, it was found that IKEA might be taking advantage of designing for the mass market based on physical and cognitive needs. This could be considered enough to be a winner in today’s market and meet the users’ needs. The design of ”KYRRE” stool is in line with IKEAs design philosophy of democratic design, that has 5 different dimensions; function, form, quality, sustainability, and low price (IKEA, n.d.). With each principle considered, a design should be considered democratic (IKEA, n.d.). It also means, according to emotional design theory (Wikberg Nilsson et al., 2021), that currently IKEA is not making an effective design of the IKEA stool ”KYRRE”, as meeting emotional needs are most likely not prioritized.The dimension of low price is working against personalisation of the product design as it drives cost when focusing on the design of the physical product. Personalisation to meet the emotional needs is left to the end user to create by themselves.

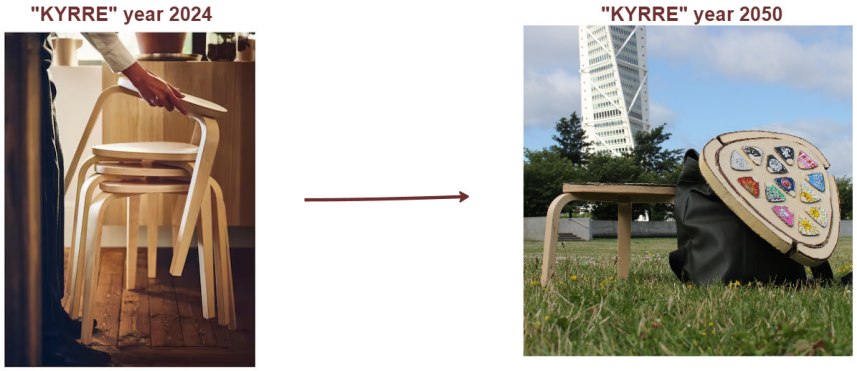


Figure 32: Current version and potentially future version of “KYRRE” (Myhrman, 2024). (Developed from image by IKEA (2024a)).

By using speculative design theory and methods, including future persona, it was possible to shift perspective and future test the design of the stool ”KYRRE” (Figure 32).

6.1. Knowledge contribution

1. Speculative design using the scenario of Copenhagen 2050 showed that sustainability drives minimalism, which means less possessions and that the product design would benefit from being multifunctional (Kang, 2021). Future persona and day-in-a-life analysis helped to further understand the young urban users need better. This was key input in the idea and prototyping phase. The result was adaption of physical form and emphasis was put on multifunctionality, foldability and that it easily could be carried around on a backpack to fit the on-the-go lifestyle of Jack.

2. The theory behind effective design, also meeting emotional needs in the design process, added new design elements to the stool with the objective of creating enjoyment as this creates positive memories and attachment to the stool (Lee Crossley, 2004). As Jack is focusing on building friendship and belonging (Maslov,1943) and living in a sustainable and; minimalistic world, the focus was on the need satisfiers Identity and Affection where embedded technology was used to add value through service to the stool.
a. The patches (personalized digital keys) with embedded technology, gives access to your personal net and with ease AI creates subjects to discuss with others.
b. The patches (personal digital keys) are personal and can be exchanged with friends and will be attached to the stool to signal identity and affection as it shows Jacks’ friends and family.

3. The final design concept of the design process was an experience mock-up prototype of the stool that was tested during the role-play method. The design concept tested showed to have an effective design, meeting physical, cognitive and emotional needs (Wikberg Nilsson et al., 2021). But, in the role-play it was also questioned if the stool was a prioritized furniture to bring with you, as part of your nomad lifestyle, as sit places in the Copenhagen 2050 scenario not is a scarce resource. Maybe it would have been better to use the backpack as the ”container” for patches instead of the stool. Presented on the outside of the backpack this would also signal Identity and Family. Here further design explorations can be done.

7. Discussion

”The purpose of this study is to contribute with increased knowledge for the design field how to utilize emotional design perspectives for future scenarios taking place in the year 2050.”

The traditional design process is often based on developing design concepts that can be launched within a short time frame to meet the high demand (Cooper, 2002). Considering user-centric design perspective means that we test the products design on existing users and they give feedback based on their current needs. This may limit important design features to be implemented that will make the product more suitable for the future. Utilizing emotional design theories to create emotional attachment is essential for the future of product design as our world is moving quickly away from ”Throw-away-society” to become a sustainable society (Cooper, 2002). With sustainability focus follows minimalism (Kang, 2021). Future product designs must be effective and thereby satisfy both physical, cognitive and emotional needs to create attachment. By using speculative design theories product designers are pushed to dramatically change their perspectives and pull themselves closer to the future user perspective. To design more user-centric in the future it is essential to have a professionally developed scenario including future persona with day-in-the-life analysis. This will give new building blocks for the design process. The final design concept needs to be further tested in a role-play where the users are briefed about the scenario, future persona and day-in-the-life to be able to use and evaluate the design concept from a future users perspective. The role-play feedback given can be summarized and then be compared with current products on the market. Are there elements we need to consider already now to be able to be competitive within 10, 15 or 20 years?

The role-play showed that the material choice was an important factor of the final design concept. The material choices were narrowed down to wood under delimitations which resulted in other materials not being further explored in this study. New sustainable materials being developed may have an impact on the perception and use of future product designs and should therefore be considered in the design process.

The role-play also showed that the users prioritized to have fewer and closer friends that they trust. Today, social media gives the users access to thousands of ”friends”. The relations are very shallow and mostly based on visual exposure. Will this change in the future and will we be more restricted with how we build relationships? This will have an impact on how we interact and what technology to use to create attachment to product design.

In the scenario Copenhagen 2050, the boundaries presented were so different vs today, that it makes it very challenging for the designer to find design opportunities for a stool. A young nomad on-the-go lifestyle means that the user have limited amount of possessions. Will the stool be a possession that they will own or will they just use the furniture being presented to them in the place they rent/stay? A new design opportunity would be to design for those offering and renting out living spaces. It could also be that the responsibility of the product life cycle is with the producer meaning that furniture is leased by users. This will further impact the design of the product.

In the scenario we also learned that there are huge differences between different generations lifestyles and needs. Will the nomad target group be big enough to drive changes in product designs and business models?

Maybe it will just be a lifestage the young users go through and then settle down prioritizing more possessions and a permanent place to live. From the study it was clear that sustainability drives minimalism and that we need to make product designs that are more effective meeting both physical, cognitive and emotional needs. Further studies can be done of how to incorporate service value into a physical product to make them more competitive in a minimalistic society. Integration of embedded technology in furniture design will most likely become more common.

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Appendix

Appendix 1: Paper booklet for participants

Welcome to the year 2050 the future of emotional design - Role Playing

August you are now Jack from the year 2050
Pär you are now Jacks new soon to be acquaintance

The scenario:

Jack is a 25-year-old student "living" in Copenhagen. He has universal connectivity via a distributed satellite-based global cloud network and uses augmented and virtual reality, voice-interactive control contact lenses to request recommendations with his AI-driven personal digital assistant (PDA), Johnny-Boy, for all aspects of his life. As a result, Jack maintains a fully on-demand lifestyle, embracing the sharing and circular economy more out of convenience and low cost rather than "green" motivations. Jack thinks about his consumption entirely within the everything-as-a-service (XaaS) space. He does not own or even rent an apartment, and only "lives" in Copenhagen when he chooses. He is able to locate living quarters or even just a bed for the night on a very short-notice basis and is never much concerned with where he might be in a day or two. Transportation, food, entertainment, and education are just as, if not more, convenient, and low-cost. Johnny-Boy monitors Jack's mood and creates hyper realistic, expensive, or immersive views, based on his understanding of Jacks' preferences. There is increasing hostility- especially among older generations - toward the tech-driven laissez-faire lifestyle of Jack and his generational cohort.

"The 15-minute city" concept has been implemented in Copenhagen as a way to reduce urban emissions, by making private motor traffic less necessary in cities, and thereby an ideal for urban planning. Residents now have an easy and quick access to all key services within 15 minutes by walking, biking, or public transport.

Disruptive technologies have significantly transformed the way urban areas as Copenhagen function. The emergence of urban tech ecosystems centered around ubiquitous sensor technology, the Internet-of-Things, coupled with cloud computing technology, and artificial intelligence, as well as mobile internet and universal connectivity, have created opportunities for increasing the efficiency optimization of urban areas and they have become "smart cities".

Giant OLED full wall screens- across all walls, ceilings, and floors - producing "natural" light have interrupted the living space market in urban cities in Copenhagen. Windows are now optimal, and this has enabled extremely high-density living spaces in urban areas, and basement apartments are no longer differentiated from penthouse with waterfront views. Even underground living - using converted underground parking spaces, refreshed by "natural" purified air - is now in Vogue as attractive locations to "spend some time".

About the stool:

- It's a portable stool with badges that are detachable.
- The stool when unfolded is a normal stool but with shorter legs
- The stool when folded can be attached to a backpack with a hook you can find on the back of the stool.
- The stool has badges that can be removed and replaced with extra once stored in the back for unexpected interactions.
- These badges are personally modified to your liking and has indents that make it possible for you as a user to scan the badge with your AR/ VR contact lenses. When the badge is scanned you will be able to see all the persons information that you met that day. Where they are, what they are doing who they are and so on. (this study and method will not further explore this feature further but its encouraged to be further explored in the future)

Appendix

Participants privacy:

- We will ensure all participants are fully informed about the study and provide their consent. This includes understanding what will be recorded and how the data will be used.
- We will ensure that the participants have access to refreshments and breaks.
- We will also ensure that all data collected is securely stored and anonymized if necessary.

Preparations:

- **Dresscode:** For this Role-Playing we are going for a futuristic look so we are thinking black shirt, and jeans or other black pants,
- **On the spot:** You will be provided with a backpack with a stool as well as a coffee cup with coffee of course :).

Estimated time it will take:

- **The schedule for the method:**
  - **Briefing (30 min)**
  - **Location 1: walking path (45-1 hour)**
  - **Break : 15 minutes**
  - **Location 2: Park (1-1,5 hours)**
  - **Debriefing: (30 min)**
  - **Feedback: (1 hour)**
  - **Total: 3-4 hours**

Script for Role-Playing:

- **Location 1: Walking path**
  - Jack walks a long distance, occasionally checking directions using AR lenses.
  - Jack stops to buy coffee.
- **Location 2: park**
  - Jack finds a spot unfolds the stool, and sits down
  - Pär approach and sit nearby.
  - **Example of Dialogue:**
    - Jack: "Hi there, enjoy the park?"
    - Pär: "Yes, it's a beautiful day. I see you have one of those new stools"
    - Jack: "Yes,It's really convenient for short breaks. "
  - **Interaction continues until one has to leave.**
  - **Badge Exchange**
    - Jack: "Here's my contact, let's stay in touch."
    - Pär: "Sure, he's mine."
- **Feedback time with questions!**

Appendix

Appendix 2: Guided questions & desk research answers

1.What makes KYRRE different from Ikeas other stools? What was the purpose of designing this specific stool?
KYRRE is an updated version of the traditional “mjölkpallen” that Ingvar Kamprad designed year 2004 (IKEA, 2024). His intention was to design a stool that was inspired by his first job which was milking cows. He found that the three legged stool didn’t wobble when placed on an uneven surface. Picture (Auctionet,2024)

Mikael Axelsson is one of the product designers at IKEA of Sweden that designed the new updated version of the KYRRE stool (IKEA, 2024). He expressed how they discovered in the designing process of KYRRE that they could optimize material usage and reduce waste by making the seat triangular instead of round. This gave the KYRRE stool its unique triangular design to make it more sustainable.

2.What makes KYRRE multifunctional was that its intended use or just happened to turn out that way?
The stool is advertised as multifunctional (IKEA,2024). The functions are sitting and or as a table. They mention and showcase on their website that the KYRRE stool is supposed to be used when unexpected guests come when you lack space. They also showcase it as a side table for a plant or a beverage.

3. For what target group were KYRRE designed for? How does that show in the design and in the options available?
By looking at how IKEA advertises the product as well as looking at Social media and in the reviews on their website I would conclude that it’s designed for a younger generation that live in small spaces (IKEA,2024).

4.Where and how is the intended use of this stool? Where in the apartment/ house? How does this show?
According to the reviews left on IKEAS website, people use it in a lot of different rooms such as the bathroom, kitchen, bedroom and living room (IKEA, 2024) .This is also displayed on their website by showcasing how different instagram users choose to incorporate the stools into their apartment. They also show live pictures on how they intend the product to be used, which is a small group of people in a small apartment.

5. How are you built/designed with IKEA’S vision and values in mind?
One of IKEA’S values is Cost-consciousness (IKEA,2024). They want to constantly challenge themself by creating affordable furniture without compensating the quality of the product. The KYRRE stool currently costs 199 SEK (IKEA,2024). This stool is one of the cheaper stools that IKEA offers.

KYRRE is a good example of how IKEA showcases simplicity. The design is simple and the design choices easily explainable: to minimize waste and to offer stability.

IKEA Showcased how they renew and improve their furniture by redesigning the previous “mjölkpallen” into the existing “KYRRE” stool. They trusted this design more by removing the previous one from the existing collection.

IKEA also shows that they give and take responsibility by telling the customer how to upkeep the chair and how to recycle it.

IKEA sustainability view 3 materials are they all necessary
The KYRRE stool fulfills the requirements of safety, sustainability and stability according to the following standard: EN 16139-Nivå 1 and ANSI/BIFMA x5.1 (IKEA, 2024). The stool’s main material is sourced from berch. IKEA claims that 98% of the wood that they use is Either FSC-certified or recycled. IKEA recommends its users to recycle the stool at a nearby recycling station.

Positive Forest Agenda 2030 is IKEA’S new goal which is to further improve their climate footprint as well as design products that can be reused, renovated, and recycled.

Mikael Axelsson a product designer at IKEA Sweden designed products with circular design in mind from the beginning stages (IKEA, 2024).

“När vi gjorde pallen KYRRE upptäckte vi till exempel att vi kunde få ut mycket mer material och minska träspill genom att göra sitsen trekantig istället för rund. Jag vill göra så få nya produkter som möjligt samtidigt som jag vill fortsätta att lösa problem som IKEA kunderna vill ha hjälp med” - Mikael Axelsson (IKEA, 2024)

6. Size, design and color, What purpose or just because of design.
The design of KYRRE is meant to both be efficient and aesthetic (IKEA, 2024). Ikea’s own words are “modern expression” and “practical help”.

7. When was it released and how was it improved from the last stools.
“KYRRE” was realized 2019 and was a updated version of Ingvar Kamprads “Mjölkpallen” that was released in 2004 (IKEA, 2024). Some improvements include the shape of the seat, which was made triangular to minimize waste. There’s a lack of Information about this 2004 “mjölkpallen” but from the picture we can see that it has significantly thinner legs.

Ikea mentions that the 3-legged stool creates significantly more stability on uneven surfaces than a 4-legged stool. Furthermore they also mention that the stools are stackable and that they don’t take up a lot of space.

Appendix

Appendix 3: Future scenario template

FUEL4DESIGN
Co-funded by the Erasmus+ Programme of the European Union

FUTURES DESIGN TOOLKIT

06 | SCENARIO

TIPS AND INSTRUCTIONS

SCENARIO

SCENARIO is a powerful tool to envision possible futures. In this stage you should select one of the scenarios developed in the previous phase then develop it into a comprehensive scenario.

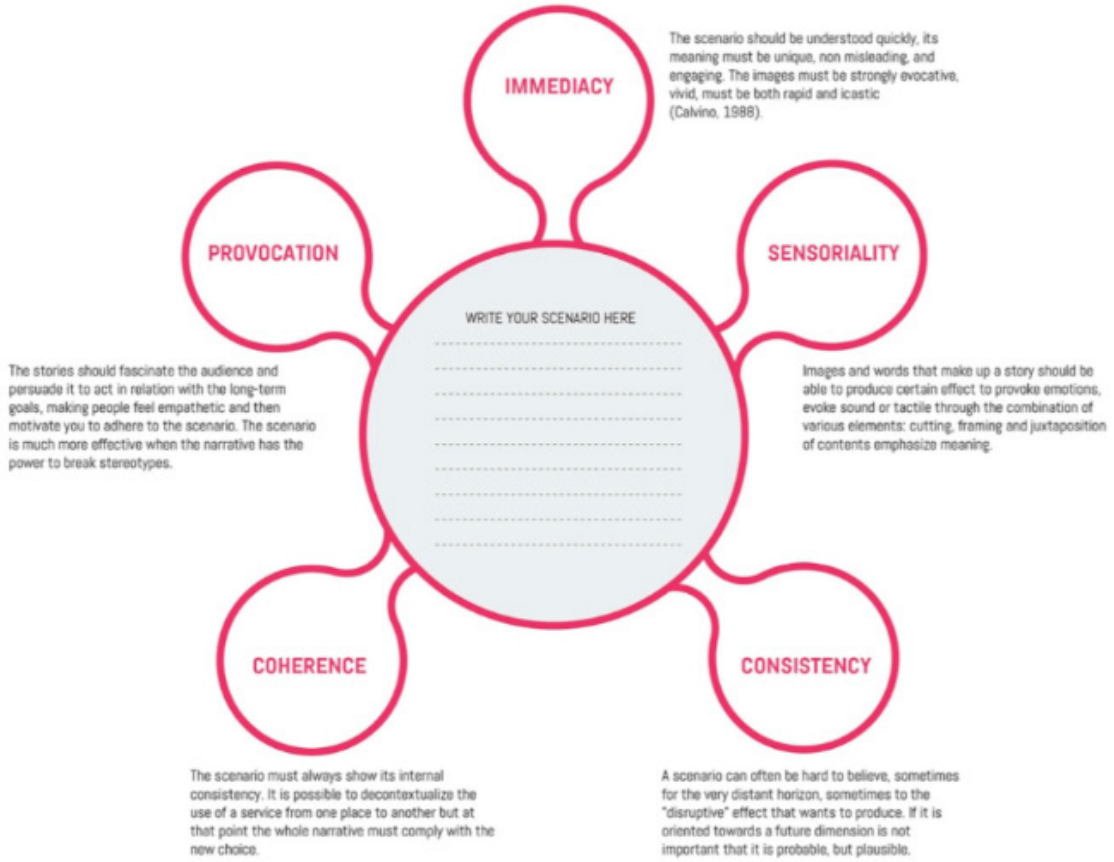
01 Quickly analyze the main elements that emerged from the scenario you've developed

02 Read the categories carefully in the scenario making template. Discuss them with your team and then situate your ideas in each category.

03 Use visual aiding material including, images, videos and sketches to describe intangible meanings and directions.

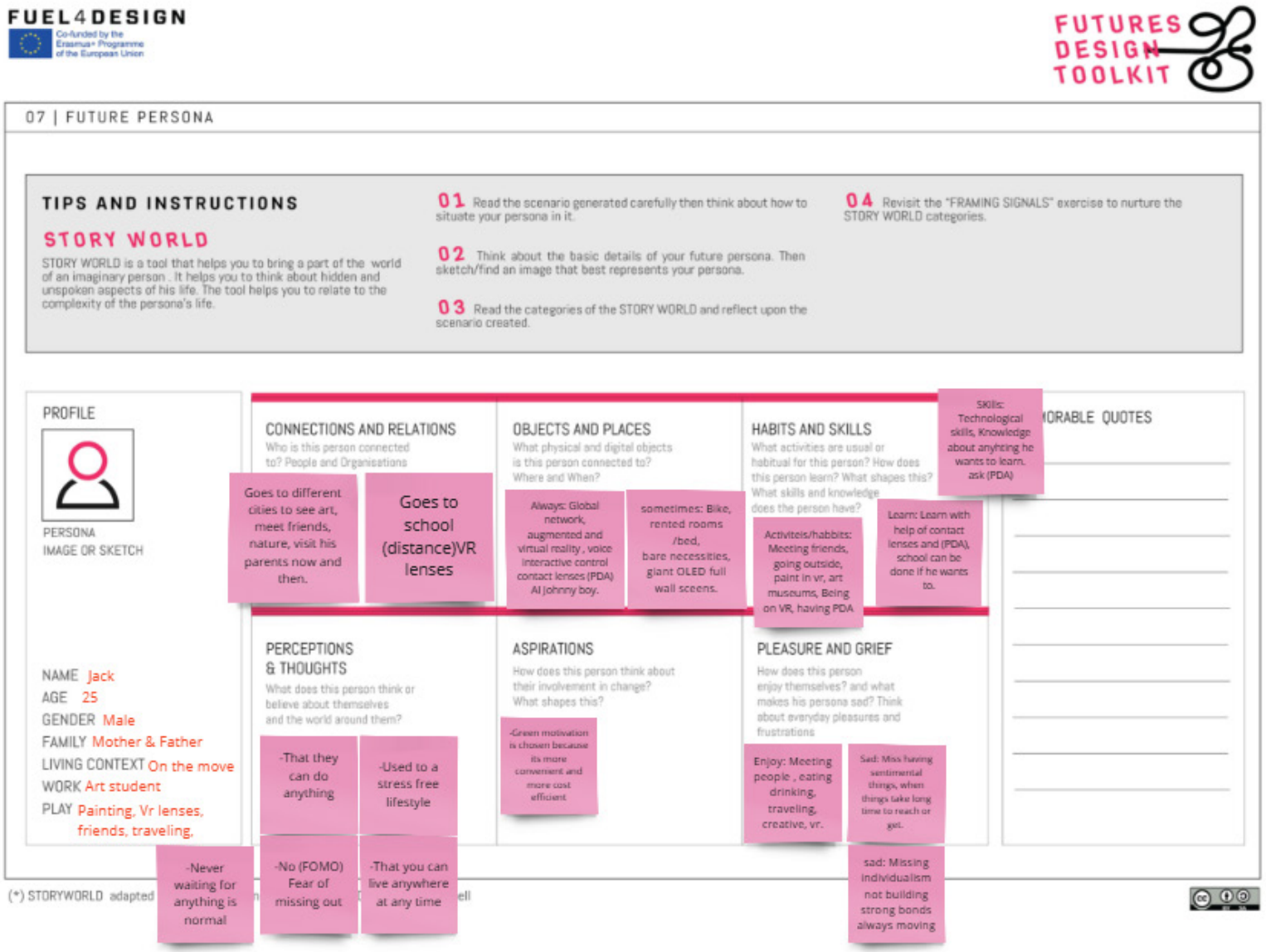
04 Consider social, cultural and technological contexts of the world you're building.

05 Write your scenario in the middle of the template.



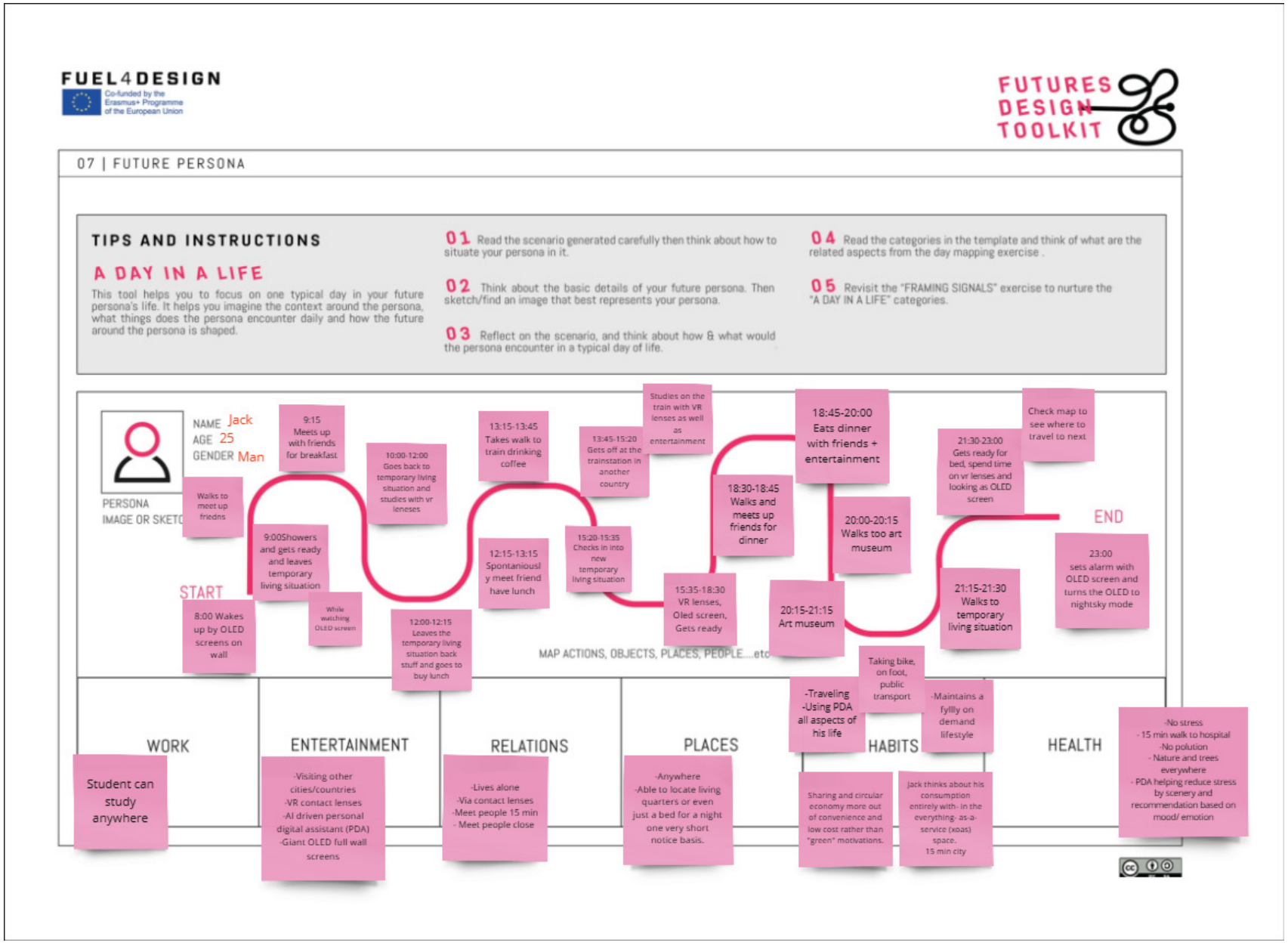
Appendix

Appendix 4: Filled out story world future persona template



Appendix

Appendix 5: Filled out a day in a life future persona template



Appendix

Appendix 6: Transcription from Role-Play

Transcription:
J: Jack, Participant 1
K: Ken, Participant 2

Transcriptions: the dialogue
J: Hello, hi!
K: Hi Hello!
J: Hello my name is Jack
K: My name is Ken, nice to see you.
J: Nice to meet you, you have one of those cool stools too
K: Yes, I cant live without one.
J: No
K: Yes, it's the only thing I have with me.
J: Yes, but I only live in the small rooms that i rent so this is the only thing I bring with me when i move.
K: Yes, I see that you have the cool tags, and I think i might see that we have some in common.
J: Yes, you were also in Belgium 2037.
K: Yes
J: Okay
K: Yes is was really awesome it was one of those cybergames.
J: Yes, you ate one of those fries with sauce on it.
K: Yes, it is like that when you look back in time, it was so trendy with retro food.
J: Yes but where else have you been?
K: No, but then I have also been to Amsterdam and had really cool room with a lot of screens, that can take you anywhere you want to be, it was there I also met Cindy.
J: Yes, I can see that from your tags.
K: Yes, but these are my best friends I am bringing with me, I can see that you also have some maybe we have some in common.
J: Yes let me check do you also have Liselott?
K: Yes, but where did you meet here?
J: Yeah, I met her in New York.
J: She was there running a marathon, do you also run marathons?
K: Yes, I also run marathons.
J: Yes, but thats awesome, do you want to run a marathon together sometime in the future?
K: Yes, but should we echange tags then?
J: Yes, I actually have one more space left here.
K: Oh, I also have one left here.
J: Me and Miss Lee are no longer friends.
K: Oh thats unfortunate, but here you have my tag.
J: Thank you, and here is mine.
K: Yes, but that's perfect, then I will look into and scan it later, thank you for getting your trust.
J: Yes you too.
K: Yes,because the world is pretty tough right now, its good to have someone you can put a little more trust in.
J: Yes, thats how it is, I can see now when im scanning that you have been to so many cool places. How was Namibia two years ago?
K: Yes, a lot has happend since then.
J: Oh, is that so?
K: Yes, but if you went back 50 years ago it was nothing like today. A lot of development has happened and now you dont have to bring a lot of things with you like you had to do before.
J: Yes big moving trucks were needed when you moved.
K: Yes the society has really gotten better its so important with relations and memories.
J: Yes and everything you need is just a few minutes away anyway so you dont need to have everything in your home its really practical.
K: I also think its so cool that you can save the memories with the tags it's so easy to remember what you have done like traveling to Namibia, running marathon and things like that.

Appendix

Appendix 7: Interview Questions and Answers from Interview Role-Play

Interview: with participants 1 and 2
J: Jack, Participant 1
K: Ken, Participant 2
How did you find the overall role-play experience?
J: It it's a little tricky to get into the role but after a while you could see what advantages this design had. It made it easy to start a conversation.
K: I think when you eventually get into the role, it then depends on what partner you have a conversation with, because when the conversation is going I think things are going pretty smoothly.
Did you find the chosen scenario immersive and realistic?
J: I can imagine that in a future when everyone own this, it becomes more realistic. Now its pretty unfamiliar, that you would approach someone with a stool in our time right now. But I can imagine this concept existing in this specific future.
K: Then I believe that there also will be more time for meeting up with people because you dont need to take care of a lot of stuff, you don't need to move a lot of things, you dont need to look for a lot of things. When you have everyhting with you there is more time to meet and have conversations with people creating memories.
Was the product easy to use in the chosen scenario?
J: Yes I do think so, it was very easy to unfold and fold and the system with exchanging contact details was easy. No, but it feels like it allued to technology that already exist today that people are used to working for example buisness cards in Japan, or that you add someone on Facebook, those things that already exist in this design a little.
K: I think it was a conversationstarter, that it is good because you can see their tags to and that makes it easier to start a conversation. For example "Oh you have been there" or " Ah that was that". That its cool when you scan the patch directly so that you can start the conversation instantly is really cool.
What difficulties did you encountered while using the product?
J: When you are sitting on the stool and having a conversation with someone it's a little bit awkward to stand up and show your tags that you just sat on. Maybe it would have been better if you could have access to them during the conversation. Maybe that you can access them on the legs of the stool or something. Now its more like i stand up and see that i traveled to Namibia and then your right buttcheek were right on it. It is definately working smootly with folding and unfolding the stool at places where you can not access seating areas. There are usually seating areas but if there werent this would work smoothtly.
K: I think that maybe it would be better if it was lighter, a material that is more futuristic and compostiple and more sustainable, but still that the tags would be on it. You could also do that, that now its a stool you sit on but you can also use it potentially as a table, it can be just as useful as a table. If you sit on the side of the stool like I did during the role-play it was easier to have a conversation and you could have the coffee of the stool. Then you can talk and not needing to sit on it. You could have your elbow on the stool too when using it as a small table.
How natural were the features of the product?
J: It is difficult to get into this whole technichal aspect, this with exchanging the tags is easy, it is a little similar to giving away a buisness card, or to add someone on a social networking site.
K: That is the thing that its like two worlds instantly connect by scanning and then isntantly seeing what you have in common like travel destinations, friends and interest, it makes it really easy to start a conversation with someone.
J: It could have been an added feature that automatically you could see what the two people have in common when you scan the patches. It could even be AI-generated. It would be perfect on a boring dinner because you can just scan the tags and have something to talk about.
K: That would have been so cool.
In what way did the product meet or fail to meet your expectations?
J: I think it met my expectations that it was really easy to use, it was handy in this scenario when you dont own a lot and have your whole home on your back.
K: There is a few things that you bring with you, it is like some clothes, and something extra. And if you need something extra you are missing you can have conversations and create memories that can help you get that homey feeling.
K: It feels weird to just have a stool with you because you are so used to having so many more things with you. To just have one thing feels really weird.
J: And that that specific thing would be a stool, feels weird because it is not really the most usefull thing out there. Because you can find places to sit in a city, lite busstops, parks, cafee sp inte not very often maybe that you have to sit anywhere else.

Appendix

How was the interaction with the other participant?
J: Yes but it does feel a little unnatural.
K: But in this kind of scenario maybe real actors would have been better to use.
J: Yes but maybe you could just meet people and connect with them through the lences that the persona has, but maybe that would be to artidficial.
In what way do you think this product can be improved?
K: Lighter material, material that exist then but maybe not right now made out of recyclable things and that is lighter.
J: Maybe Alluminium. Patches can be recyclable if people exchange them often. I think the design is nice and I like that the shape of the seat matches the shape of the patches it looks nice. An i like how it goes from 2D to 3D design when unfoldin-git and folding it. The legs give the design better support.
What would you say can be improved about this role-playing method execu-tion for future studies?
J: In more specifik situations, that we get those promts now, it was about "time to talk about this chair now", maybe it would be better if you said "Okay you are going to exchange contact details, here are the patches, change them with each other", or " Now you are waiting for the bus, there is no bench continue". or "Now you are standing in a grocery store, the queue is very long, what do you do?".
K: Yes, to be put into a situation is good, but I think it is cool to have to figure out a creaitve way to solve this thing, if you have visualized something, set up a scenario and then put yourself into that scenario, in that situation and think about what to do now and new ideas are born. But I liked the idea participant 1(Jack) had to scan the patch directly when you meet and you can see everything you have in common, that they meet in a bigger perspective than like on Facebook today. Then its so far way and you do not really know who the person is, and there are few good things to come out from it. In this situations you give your trust and suddenly you will find out 5 things you are interested in, where you can contribute, you can have more fun and effective conversations this way.
How did the use of AR/VR lenses and AI influence your experience with the design and the scenario?
J: It made it easy to get to know eachother really quickly by having access to their information. The key is the patch, it was enough to just look at the coffee with my lences to be able to buy it.

Other:
J: A lot of parks gets you motivated to move more outdoors, and then this design will be useful. And because everything is accessable within 15 minutes it also makes it easier to spend less time at home.
K: What if the person grows because "who I am" is now more important than "what I own". In the past it was more about "what kind of car I have", "where do I live", "what house do I have". The shift in perspective makes it more interesting.
Do you think you can eventually feel more for this stool than a stool you own today?
J: As you meet more people and collect their stories on the chair, I think that yes. But then the chair doesn't become a chair, but also the people behind it. But today everything is pretty serile that a chair is a chair and wood and metal. But now you can have your friends in your living room when you drink your coffee and put it down on the table and see, right here is the person I met. Awakens a stronger emo-tional bond, I've had that with me on my back,
K: The other furniture today only fulfills ne function, now it will be something I have with me in conversation starters and such. Giving up all the other furniture I need something that is light in weight, but still visible, it's almost like a photo album, it was fun we did there, you can scan again and check again, share one mine.de other furniture today only fulfills ne function, now it will be something I have with me in conversation starters and such. And if I've given up on all other furniture, I need something that is light in weight, but invisible, it's almost like a photo album, it was fun we did there, you can scan again and check again, share a mine.
Possibilites of improvement:
J: Quite well developed right now, this IT functionality is not in place and it would have been the next step, and to connect and that in an early step, to be able to have QR codes on these ships, so that you could have read on mobile, already on the market already within years, then in the future you can add technology,
K: What I was thinking about was that even back in time you had jackets like this where you sewed on badges, when I went skiing, sewed on I've been there and there, cool and A have you been there too, conversation starter, a bit the same thing, analogue now it's going to be digital, modern conversation starter.
J: There is almost another product that is missing for this product and that is a bag or carrying solution, bag that is adapted to this or some type of baby carrier.
K: It might be sitting outside my backpack which is designed on a
J/K: The same shape is the one on the back plate of the backpack, at the far end of the back, can pull it out like a laptop compartment. Wakes out. Walking with it outside the bag as it is now felt a bit cumbersome. And if you have another material that is thinner, it might be easy to just put down just as a small shield. Then maybe it's good for the back. You might even be able to fold it in two,
J: Stylish, emotional, modular (you can replace it as well as not static)
K: Relationship starter, memory bank and multifunctional

Appendix

Appendix 8: User-Centric design process

STARTING POINT

Needs and criterias affecting the design process

Maslow

Maslow	Max-Neef
Self-Esteem needs	**Identity**
Confidence, Achievements, Respect of others, Connections, Need for individuality	*Being:* Self-esteem, solidarity, respect, tolerance, generosity, passion, sense of humor *Having:* Friendship, family, partnerships, relationships with nature *Doing:* Make love, caress, express emotions, share, take care of, cultivate, appreciate *Interacting:* Privacy, intimacy, home, spaces of togetherness
Love and Belongingness needs	**Affection**
Friendship, Family, Intimacy, Connections	*Being:* Sense of belonging, consitency, self-esteem, differentiation *Having:* Symbols, language, religion, habits, values, norms, memory, work... *Doing:* Commit oneself, integrate oneself, confront, decide on, recognize oneself *Interacting:* Social rythm, everyday setting, maturation stages

Wikberg Nilsson

Functional design (We see)	Cognitive design ((We understand)	Emotional Design (We like)
Stable construction in natural materials, wood Multifunctional; stool & table		Sustainable and minimalistic concept

Starting point as of today
First round - Sketching & Cardboard prototyping based on user insights from Introduction, Background ,Theory and Desk Research
Second round - Cardboard Prototyping based on additional user insights from Scenario
Third round - Role play - additional user insights from Role play to incorporate in future design

Appendix

Appendix 8: User-Centric Design Process

FIRST ROUND Sketching & Cardboard prototyping

Needs and criterias affecting the design process

Maslow

Max Neef

Self-Esteem needs	**Identity**
Confidence, Achievements, Respect of others, Connections, Need for individuality	*Being:* Self-esteem, solidarity, respect, tolerance, generosity, passion, sense of humor *Having:* Friendship, family, partnerships, relationships with nature *Doing:* Make love, caress, express emotions, share, take care of, cultivate, appreciate *Interacting:* Privacy, intimacy, home, spaces of togetherness
Love and Belongingness needs	**Affection**
Friendship, Family, Intimacy, Connections	*Being:* Sense of belonging, consitency, self-esteem, differentiation *Having:* Symbols, language, religion, habits, values, norms, memory, work... *Doing:* Commit oneself, integrate oneself, confront, decide on, recognize oneself *Interacting:* Social rythm, everyday setting, maturation stages

Wikberg Nilsson

Functional design (We see)	**Cognitive design ((We understand)**	**Emotional Design (We like)**
Stable construction in natural materials, wood	On-the-go lifestyle - Make small	Sustainable and minimalistic concept
Multifunctional; stool & table	Circular concept by resusing parts (tested and terminated)	

Starting point as of today
First round - Sketching & Cardboard prototyping based on user insights from Introduction, Background ,Theory and Desk Research
Second round - Cardboard Prototyping based on additional user insights from Scenario
Third round - Role play - additional user insights from Role play to incorporate in future design

Appendix

Appendix 8: User-Centric Design Process

SECOND ROUND - Cardboard prototyping

Needs and criterias affecting the design process

Maslow

	Max-Neef
Self-Esteem needs	**Identity**
Confidence, Achievements, Respect of others, Connections, Need for individuality	*Being:* Self-esteem, solidarity, respect, tolerance, generosity, passion, sense of humor *Having:* Friendship, family, partnerships, relationships with nature *Doing:* Make love, caress, express emotions, share, take care of, cultivate, appreciate *Interacting:* Privacy, intimacy, home, spaces of togetherness
Love and Belongingness needs	**Affection**
Friendship, Family, Intimacy, Connections	*Being:* Sense of belonging, consitency, self-esteem, differentiation *Having:* Symbols, language, religion, habits, values, norms, memory, work... *Doing:* Commit oneself, integrate oneself, confront, decide on, recognize oneself *Interacting:* Social rythm, everyday setting, maturation stages

Wikberg Nilsson

Functional design (We see)	**Cognitive design ((We understand)**	**Emotional Design (We like)**
Stable construction in natural materials, wood Multifunctional; stool & table	On-the-go lifestyle - Make small	Sustainable and minimalistic concept Connecting people and creating memories, Embedded technology
From circular to more space efficient triangular shape of seat Foldable to use less pace	Portable to be carried on a backpack Customizable with patches to make it personal	Enhance user identity

Starting point as of today
First round - Sketching & Cardboard prototyping based on user insights from Introduction, Background ,Theory and Desk Research
Second round - Cardboard Prototyping based on additional user insights from Scenario
Third round - Role play - additional user insights from Role play to incorporate in future design

Appendix

Appendix 8: User-Centric Design Process

THIRD ROUND - Role-play

Needs and criterias affecting the design process

Maslow / Max Neef

Maslow	Max Neef
Self-Esteem needs	**Identity**
Confidence, Achievements, Respect of others, Connections, Need for individuality	*Being:* Self-esteem, solidarity, respect, tolerance, generosity, passion, sense of humor *Having:* Friendship, family, partnerships, relationships with nature *Doing:* Make love, caress, express emotions, share, take care of, cultivate, appreciate *Interacting:* Privacy, intimacy, home, spaces of togetherness
Love and Belongingness needs	**Affection**
Friendship, Family, Intimacy, Connections	*Being:* Sense of belonging, consitency, self-esteem, differentiation *Having:* Symbols, language, religion, habits, values, norms, memory, work... *Doing:* Commit oneself, integrate oneself, confront, decide on, recognize oneself *Interacting:* Social rythm, everyday setting, maturation stages

Wikberg Nilsson

Functional design (We see)	Cognitive design ((We understand)	Emotional Design (We like)
Stable construction in natural materials, wood	On-the-go lifestyle - Make small	Sustainable and minimalistic concept
Multifunctional; stool & table		Connecting people and creating memories, Embedded technology
From circular to more space efficient triangular shape of seat	Portable to be carried on a backpack	Enhance user identity
Foldable to use less pace	Customizable with patches to make it personal	Encourage social connections that create enjoyment and memories
Use lightweight more futuristic and sustainable material	Develop communication software with support of AI	

Starting point as of today
First round - Sketching & Cardboard prototyping based on user insights from Introduction, Background ,Theory and Desk Research
Second round - Cardboard Prototyping based on additional user insights from Scenario
Third round - Role play - additional user insights from Role play to incorporate in future design